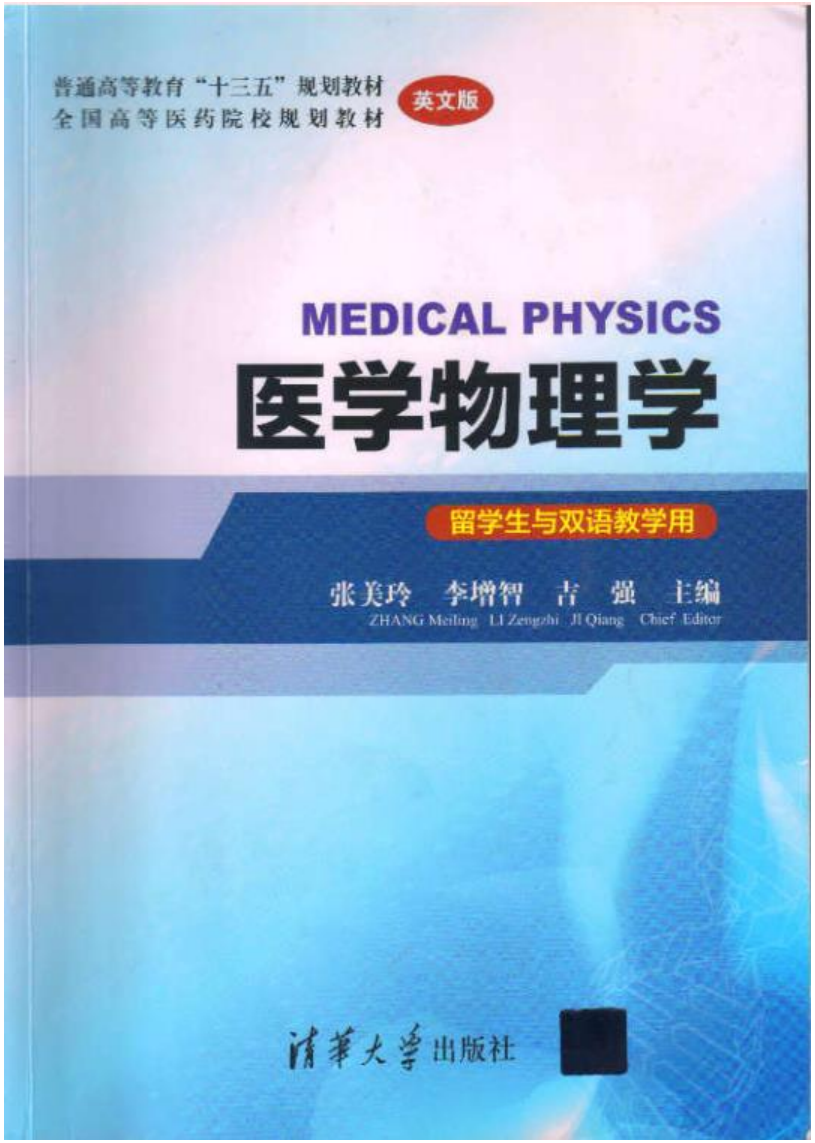
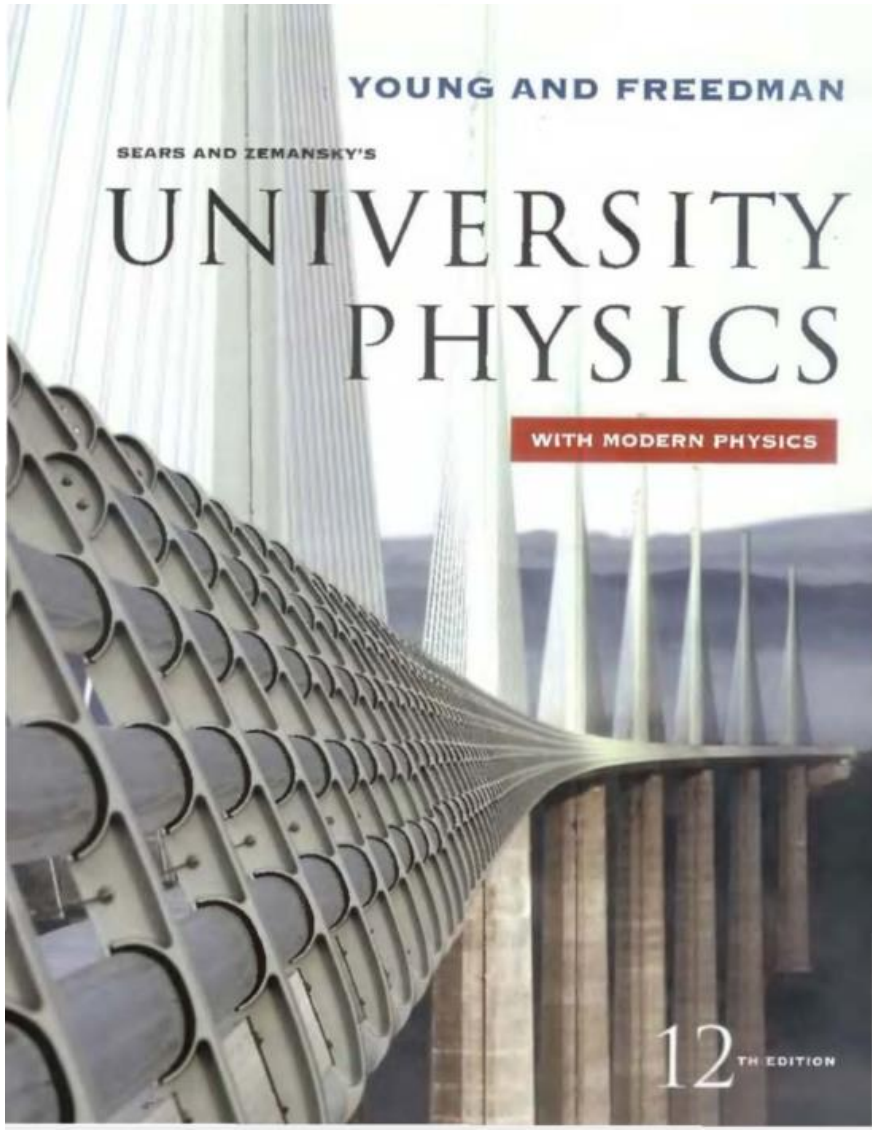
Medical Physics

- **Lecturer: Prof. WANG YE**
- **Email: wangye@zzu.edu.cn**
- **Office： Room 305, Building 5, South Campus,**
- **School of Physics**

Text book


普通高等教育“十三五”规划教材
全国高等医药院校规划教材
英文版
MEDICAL PHYSICS
医学物理学
留学生与双语教学用
张美玲 李增智 吉 强 主编
ZHANG Meiling LI Zengzhi JI Qiang Chief Editor
清華大學出版社

Main reference



Young and Freedman. University Physics[M]. 12th ed. New Jersey: Addison-Wesley. 2010

Acknowledgement

Prof. Zhang Meiling

Prof. Li Zengzhi

Prof. Ji Qiang

Marks

Attendance: 15%

Homework： 15%

QA: 10%

Final-exam: 60%

How to take class

Attendance: Sign on the attendance sheet.

Homework: pdf file, send to monitors (Iben Walid Lilid – 202380040119 for Civil Engineering 202380090148 Golam Rabbi for Software Engineering),

and monitors send to me together before the next week Monday's class. There is another chance to submit all missing homeworks to me before the final exam by Wechat.

Homework file name: Roll number+week 1 + name,

For example: 2023800901XX + week 1 + Alfa Robot

Class materials (PPT slides etc.): Wechat group

For example of Home work

Example:

Page 33:

Q10, 11， 13， 17， 24， 26

Homework file name:

Roll number+week 3 + name,

for example: 202380090199 + week 3 + Alfa Robot

In pdf file, send to monitors (Iben Walid Lilid – 202380040119 for Civil Engineering 202380090148 Golam Rabbi for Software Engineering)

Content

Chapter 1: Statics and Translations
Chapter 2: Rotation and Periodic Motions
Chapter 3: Fluids
Chapter 4: Thermodynamics
Chapter 5: Electrostatics
Chapter 6: Direct Current
Chapter 7: Magnetic Field
Chapter 8: Electromagnetic Induction
Chapter 9: Optics
Chapter 10: Special Relativity
Chapter 11: Quantum Physics

Chapter 7: Magnetic Field

Seven Fundamental Quantities in International System of Units (SI)

Quantity	SI
1. length	meters (m)
2. time	seconds (s)
3. mass	kilograms (kg)
4. temperature	kelvins (K)
5. electric current	amperes (A)
6. amount of substance	moles (mol)
7. luminous intensity	candelas (cd)

Chapter 7: Magnetic Field

Force of charged particles in magnetic field

$$F = qv \times B = qvB\sin\theta$$

$$R = \frac{mv}{qB}$$

$$\oiint_S \vec{B} \cdot d\vec{S} = 0$$

Ampere's Law

$$\int B \cdot dl = \mu_0 I$$

Magnetic forces on current

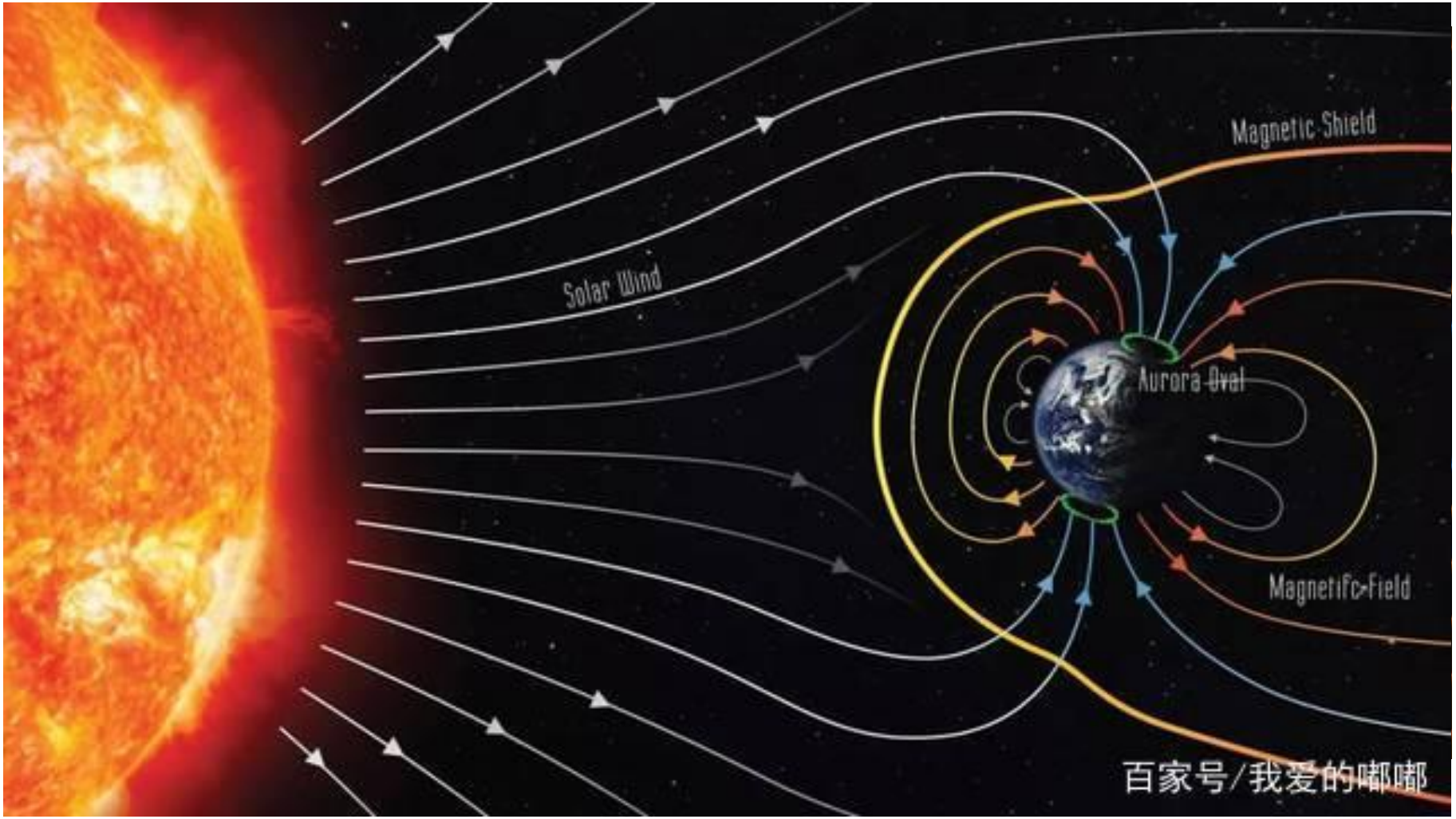
$$dF = Idl \times \mathrm{B} \qquad F = I\int dl \times B$$

Hall effect

$$eE_H = evB \qquad V_H = \frac{I}{bne}B$$

Why there is no life on Mars?

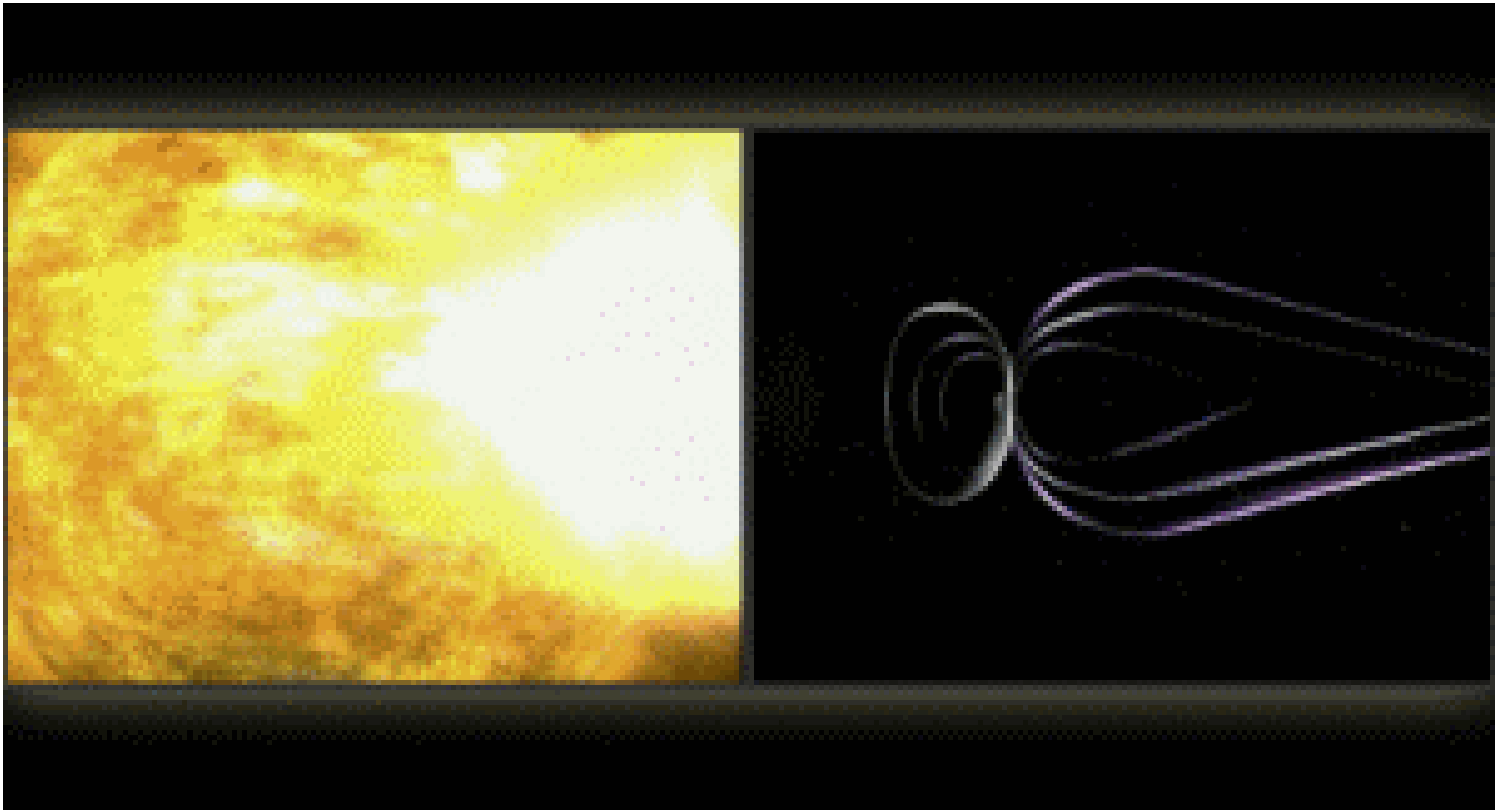
•7.1.1 Magnetic Field



Magnetic field of the earth protect us!

http://baijiahao.baidu.com/s?id=1667390531771230892&wfr=spider&for=pc

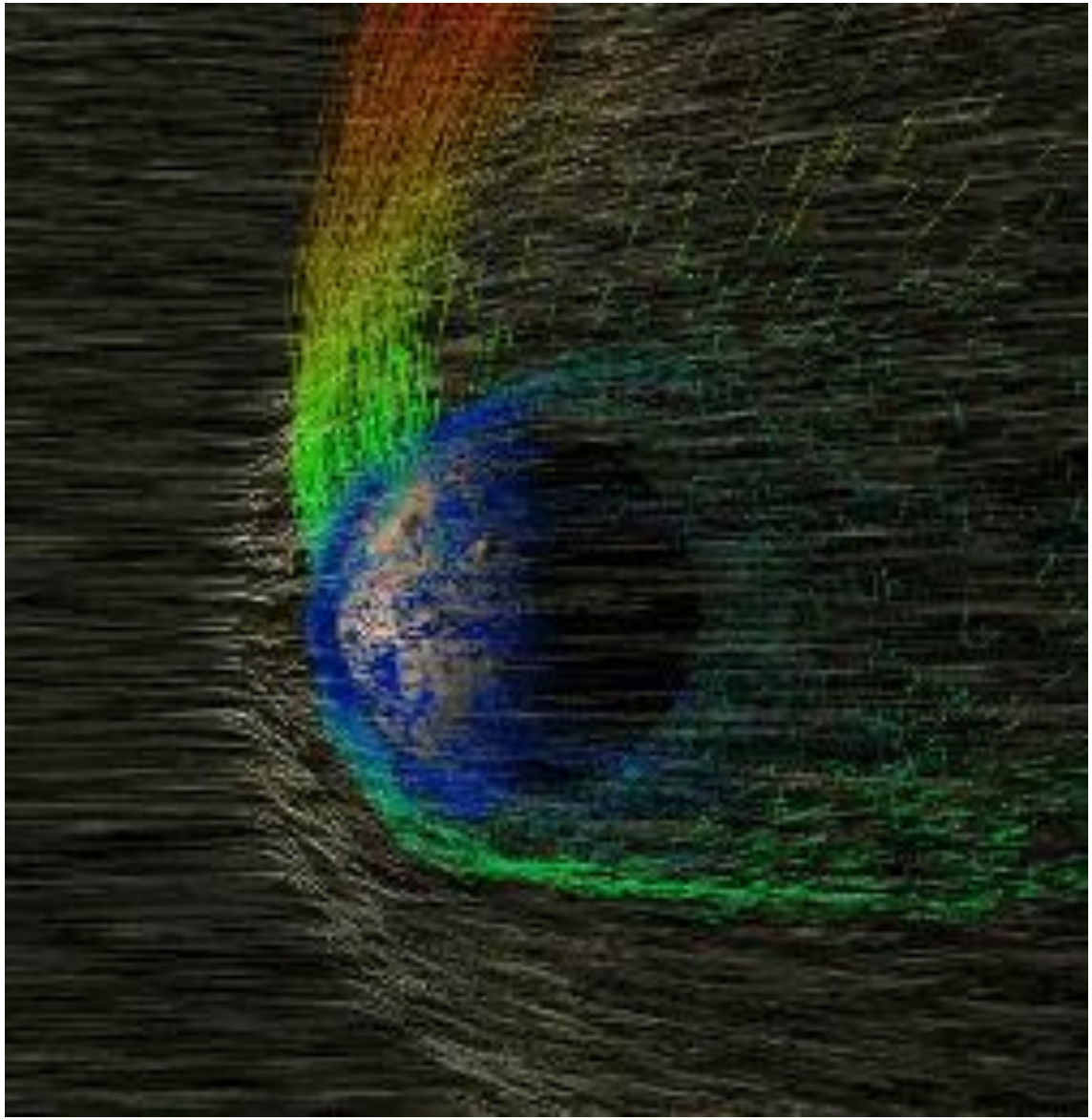
•7.1.1 Earth Magnetic Field:



Solar Prominences: solar wind

https://mbd.baidu.com/newspage/data/landingpage?context=%7B%22nid%22%3A%22news_15420796521191404167%22%2C%22sourceFrom%22%3A%22wiseindex%22%7D

7.1.1 Earth Magnetic Field



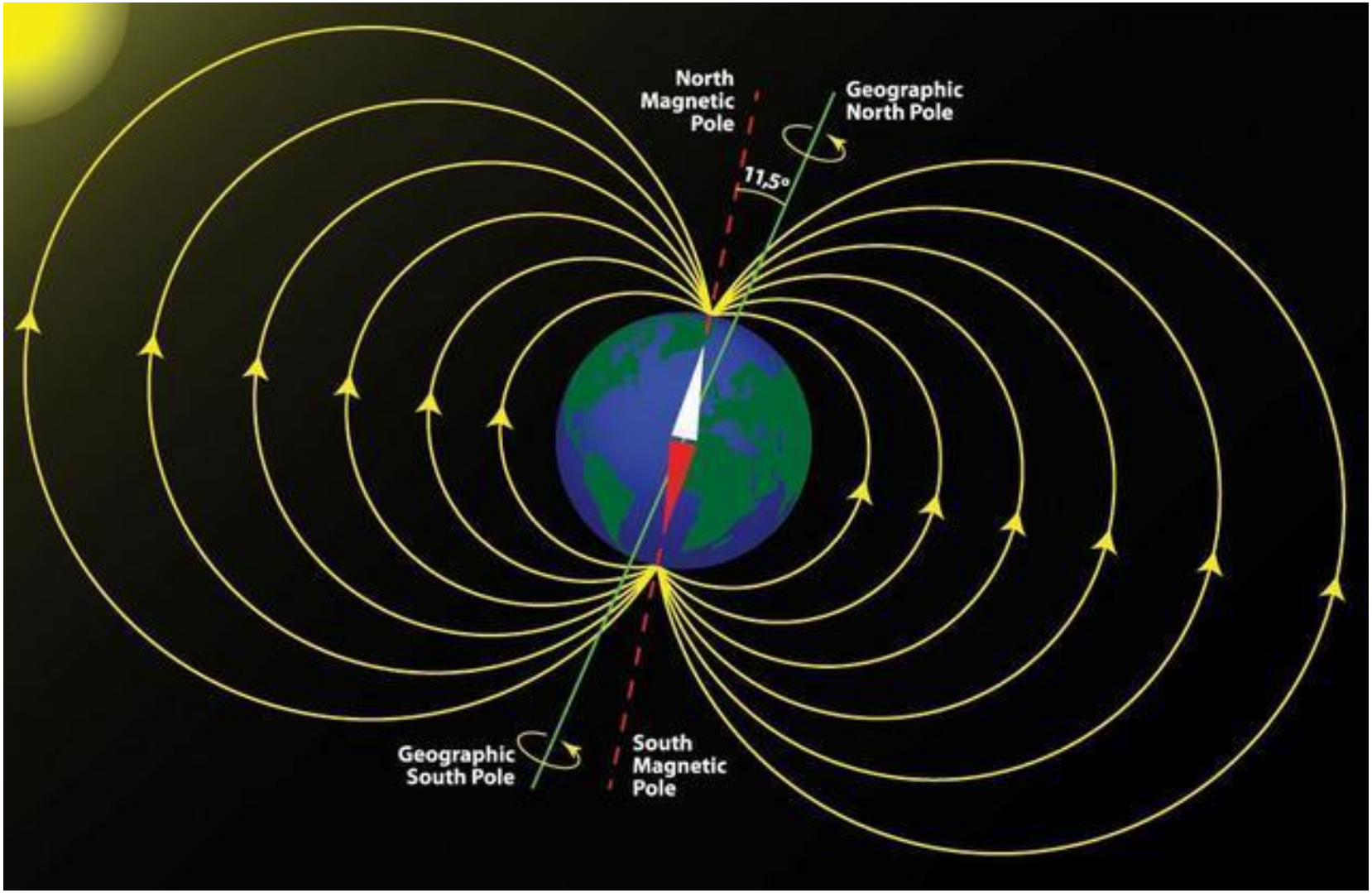
https://mbd.baidu.com/newspage/data/landingpage?context=%7B%22nid%22%3A%22news_15420796521191404167%22%2C%22sourceFrom%22%3A%22wiseindex%22%7D

•7.1.1 Earth Magnetic Field: Aurora



https://mbd.baidu.com/newspage/data/landingpage?context=%7B%22nid%22%3A%22news_15420796521191404167%22%2C%22sourceFrom%22%3A%22wiseindex%22%7D

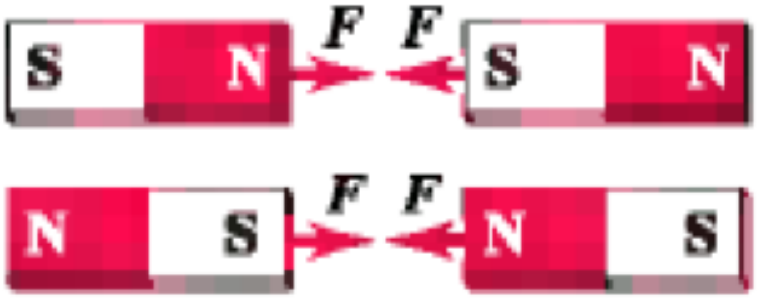
7.1.1 Earth's magnetic field



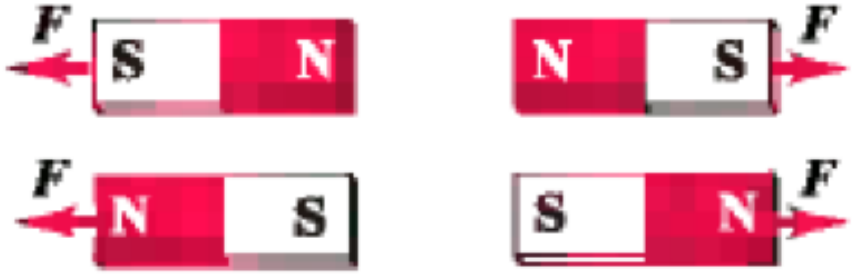
The field, which is caused by currents in the earth's molten core

7.1.1 Magnetism

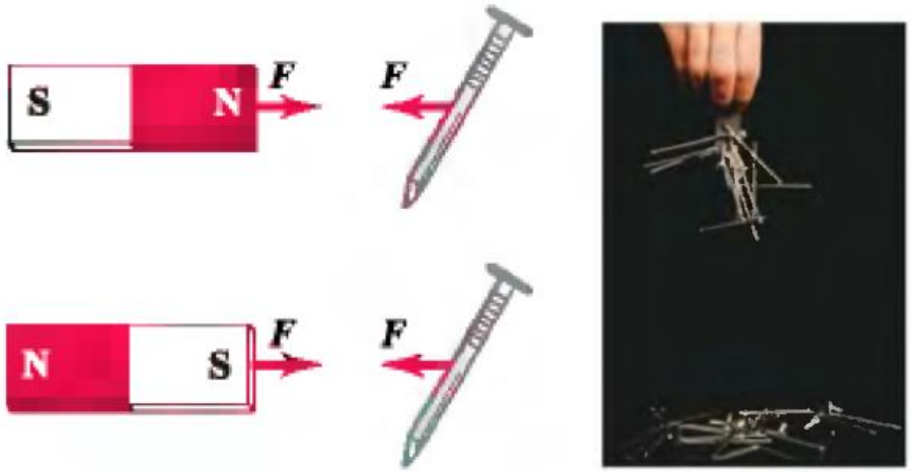
Opposite poles attract



Like poles repel

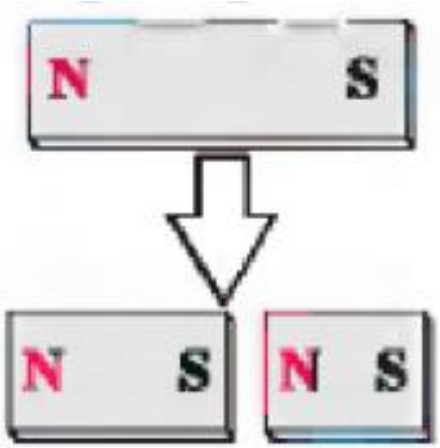


7.1.1 Magnetism



Either pole of a bar magnet attracts an unmagnetized object that contains iron, such as a nail

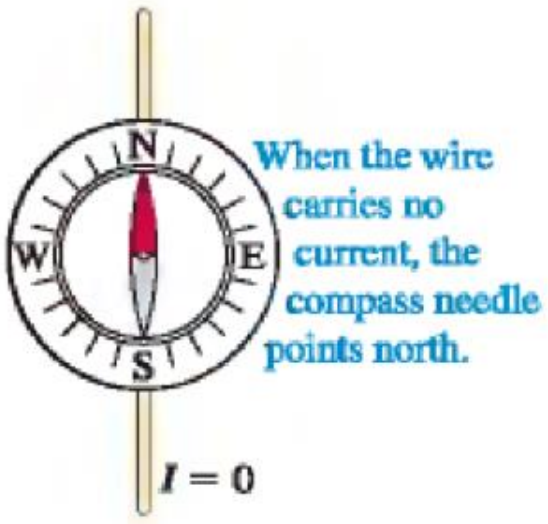
Breaking a magnet in two…



Yields two magnets, not two isolated poles

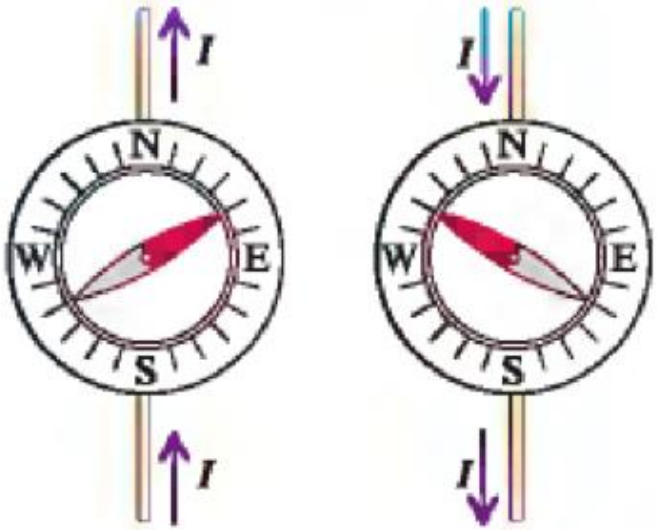
7.1.1 Magnetism

No current



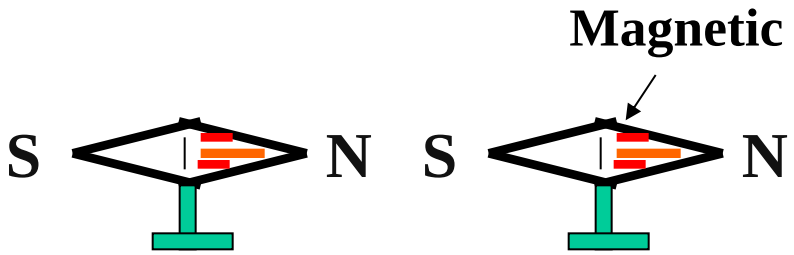
A compass is placed over a horizontal wire without current

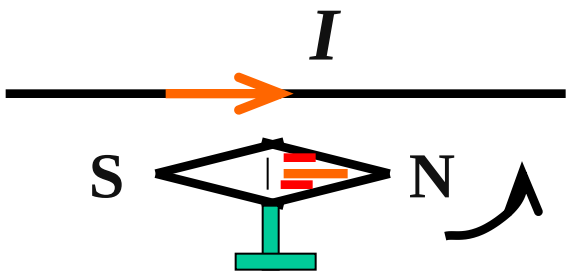
With current

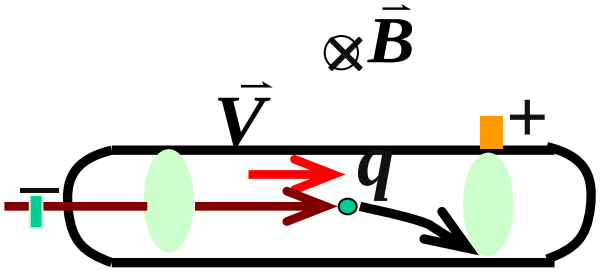


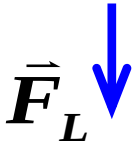
The compass is deflected when there is a current. When the current reversed, the compass deflection is also reversed.

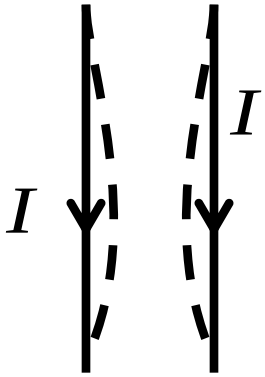
7.1.1 Magnetism







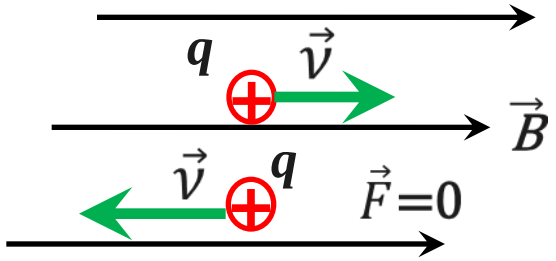




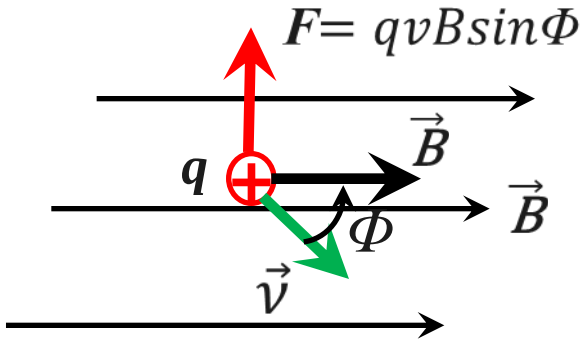
Magnetism is related with the action between the moving charges.

7.1.1 The force subjected to the charged particle

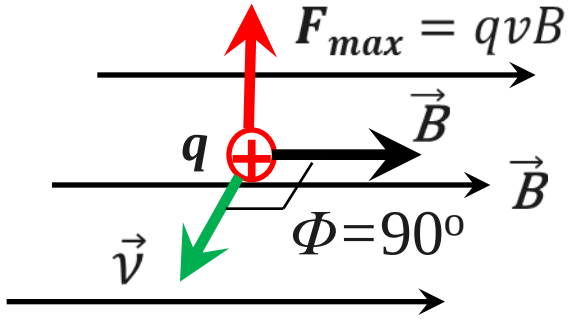
(a) The moving direction of the charge is parallel to $\vec{B}$

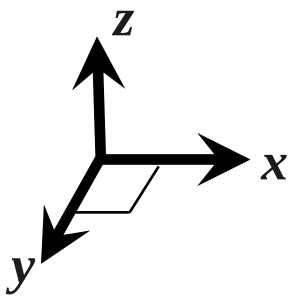


(b) A charge moving at an angle Φ *to* $\vec{B}$



(c) A charge moving perpendicular to $\vec{B}$ experiences a maximal force





7.1.1 Magnetic Field

The magnetic field exerts a force on the moving charge. This force is related to the velocity of charge:

$$\boldsymbol{F} = qv \times B = qvB\sin\Phi$$

Its direction is determined by **Right-hand Rule**

$\times$ means cross multiplication, $\sin\theta$

$\cdot$ means point multiplication, $\cos\theta$

Right-hand Rule: positive charge

Three steps:

1st step:

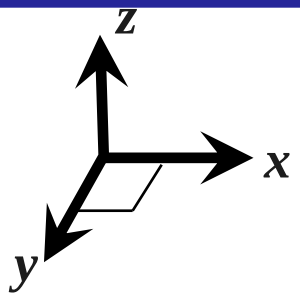
Place the $\vec{v}$ and $\vec{B}$ vectors tail to tail.

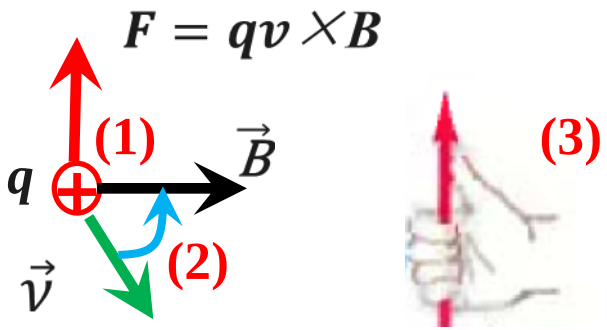
2nd step:

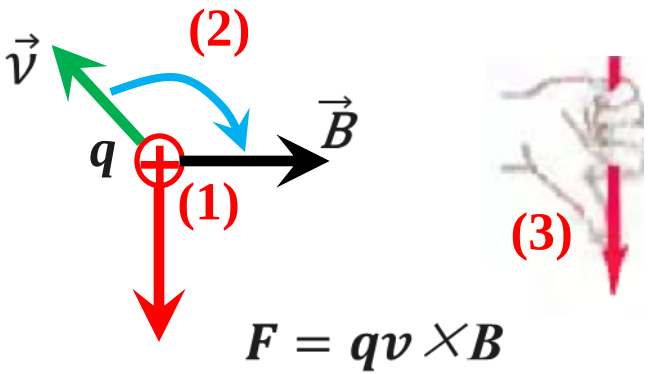
Imagine turning $\vec{v}$ toward $\vec{B}$ in the $\vec{v}$ - $\vec{B}$ plane (through the smaller angle)

3rd step:

The force acts along a line perpendicular to the $\vec{v}$ - $\vec{B}$ B plane. Curl the fingers of your right hand around this line in the same direction you rotated $\vec{v}$. Your thumb now points in the direction the force acts



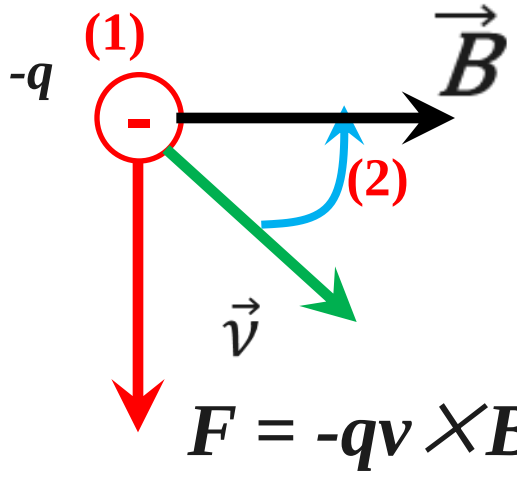


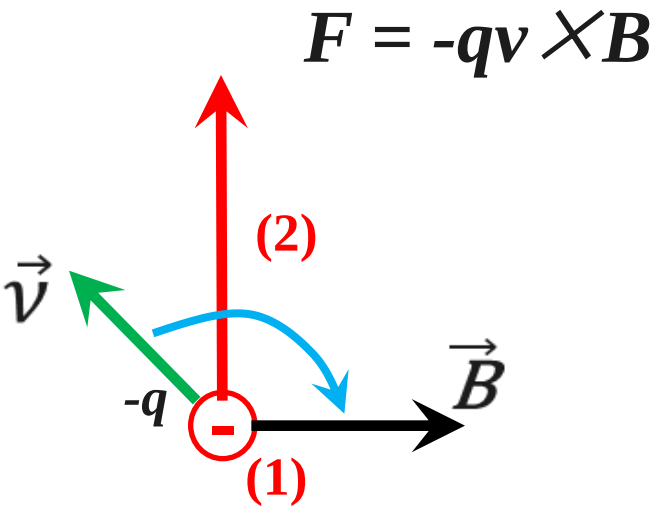


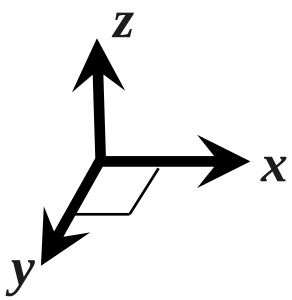
Q: what is direction for the charge is negative?

Right-hand Rule: negative charge

If the charge is negative, the direction of the force is opposite to that given by the right-hand rule

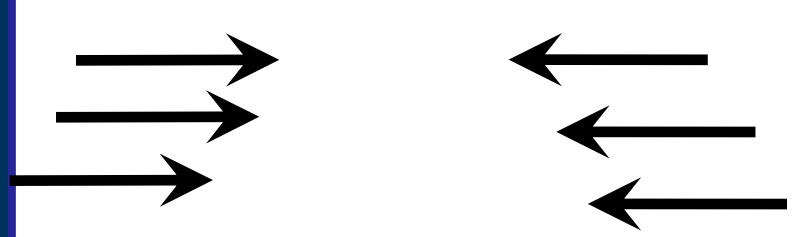






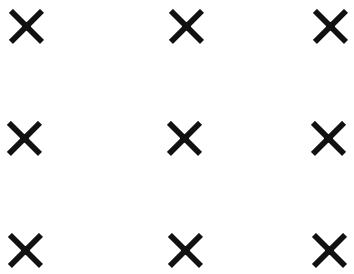
B directions

$\vec{B}$

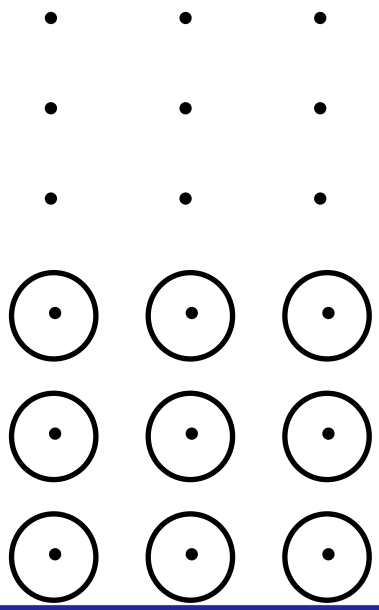


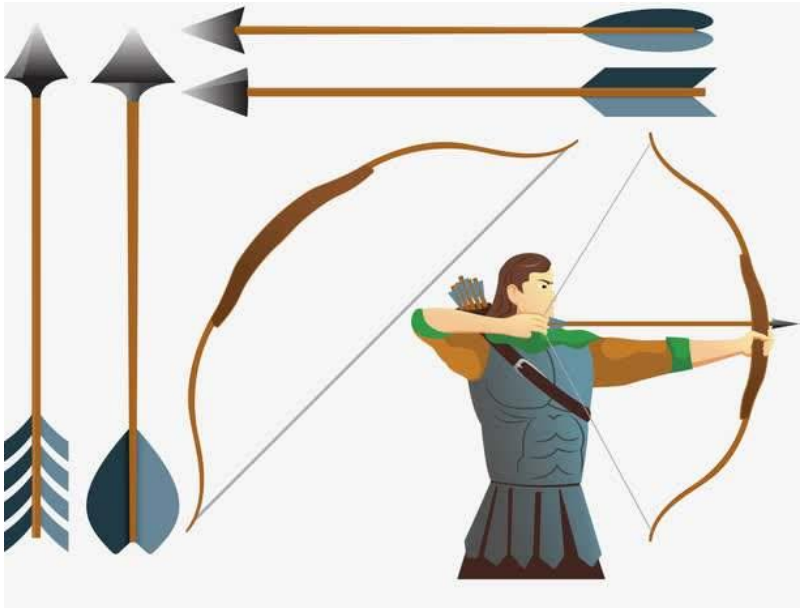


Go inside the wall/screen/paper

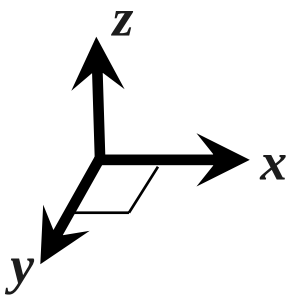


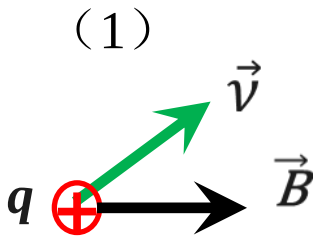
Go outside the wall/screen/paper

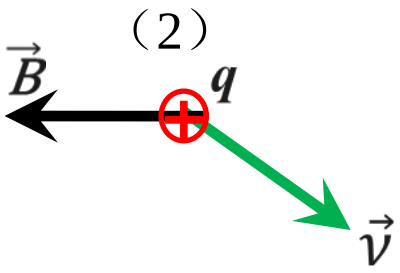


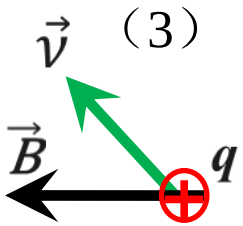


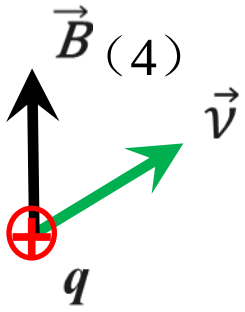
More practice for Right-hand rule

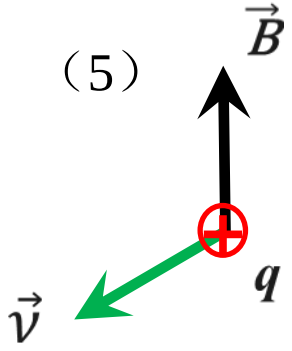


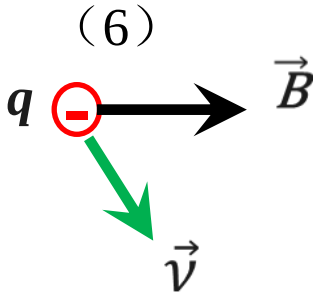


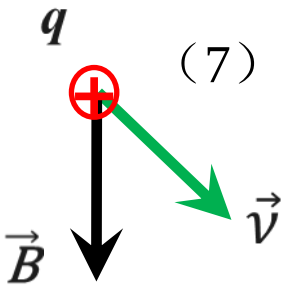


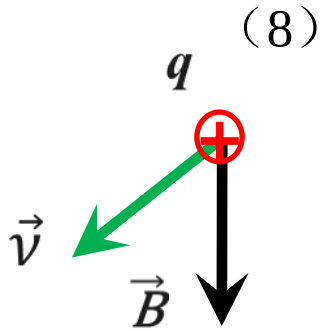




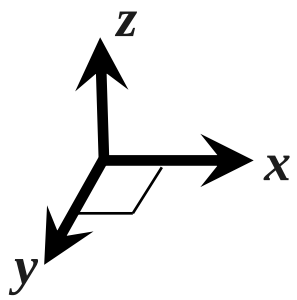


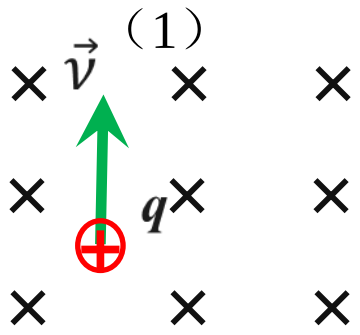


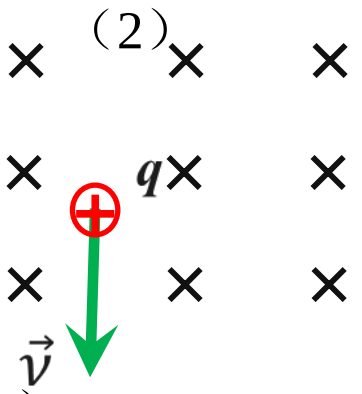


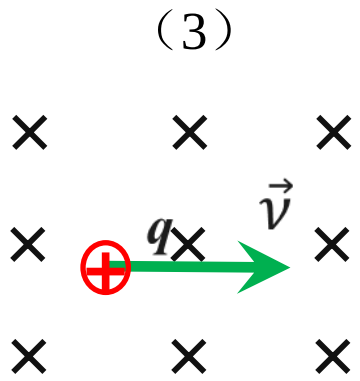


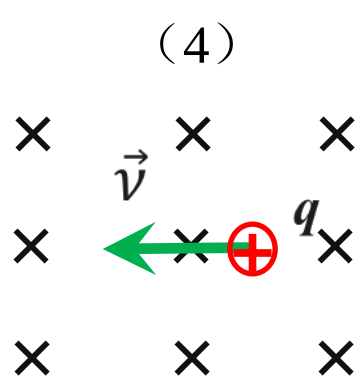
More practice for Right-hand rule

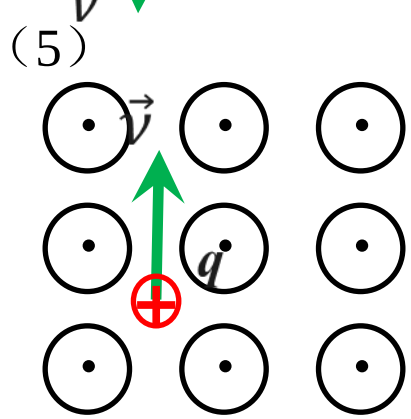


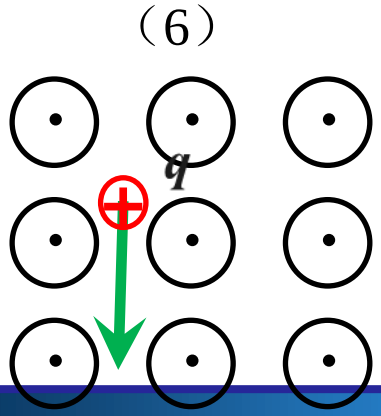


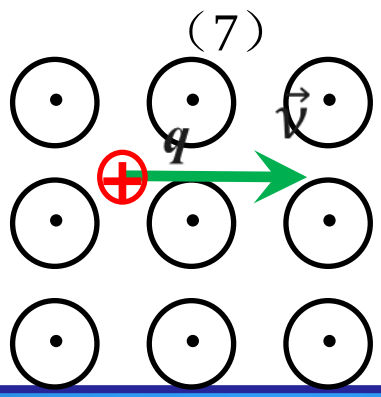


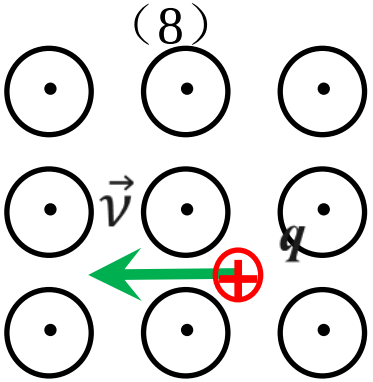






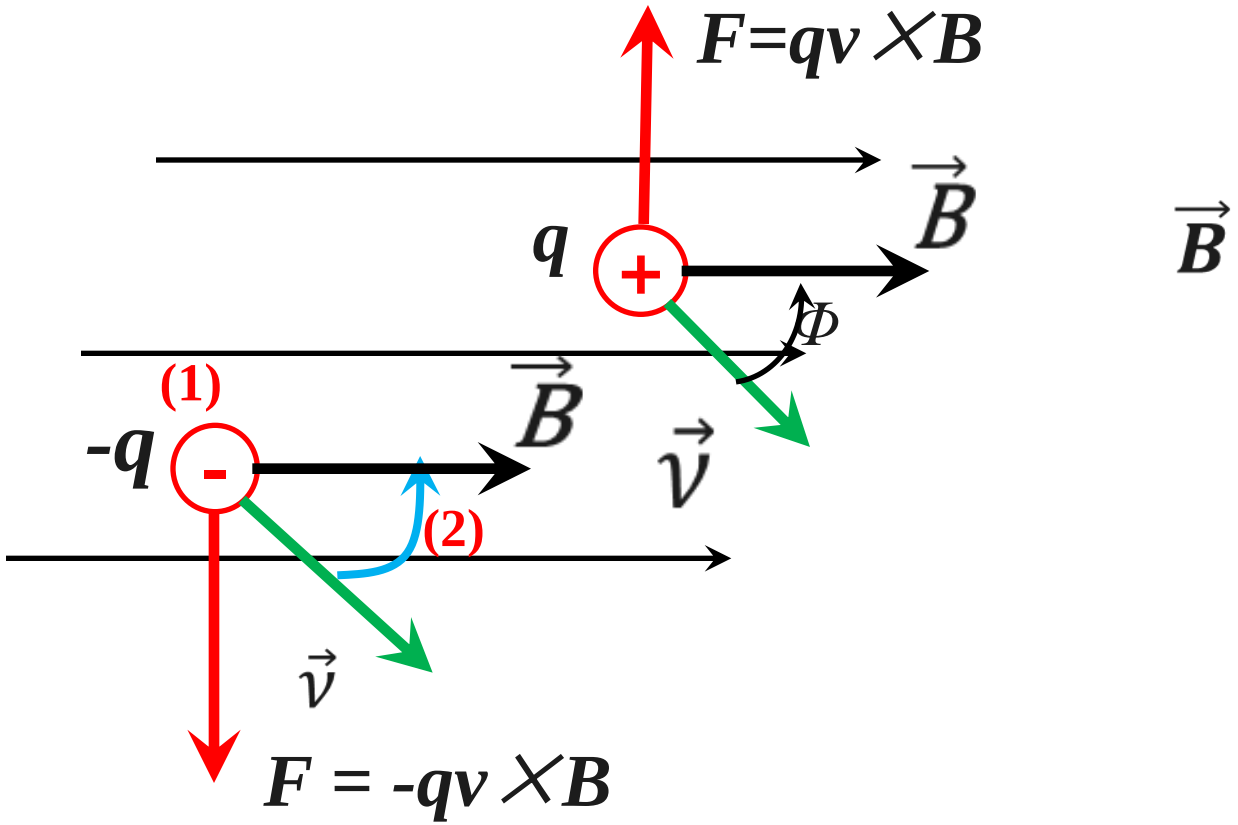




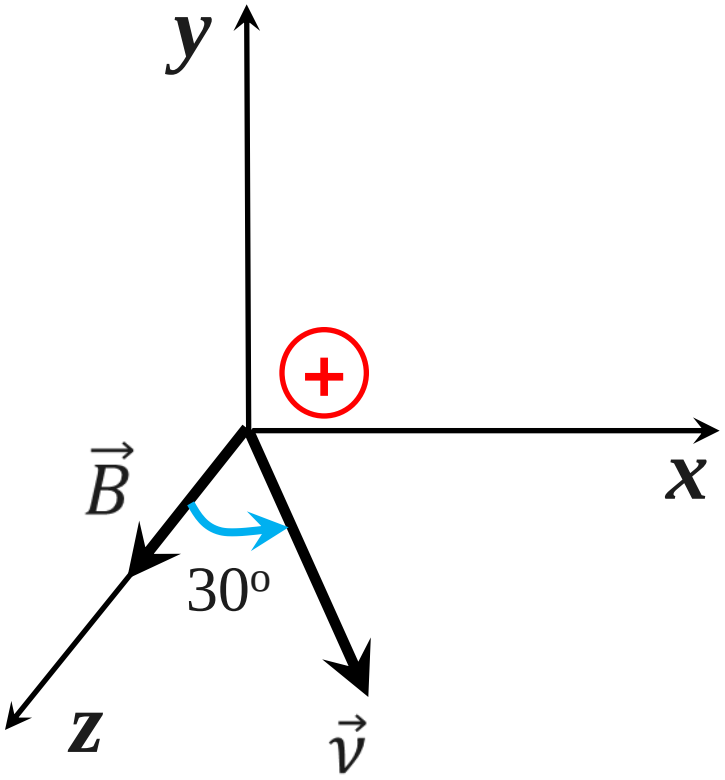


Right-hand Rule: negative charge

Positive and negative charges moving in the same direction through a magnetic field experience magnetic forces in opposite directions



An example



A proton beam moves through a region of space where there is a uniform magnetic field of magnitude 2.0T, with direction along the positive z-axis. The protons have velocity of magnitude $3 \times 10^{5} m \cdot s^{-1}$ in the xz-plane at an angle of 30° to the positive z-axis. Find the force on a proton ($q = 1.6 \times 10^{-19} C$)

Solution

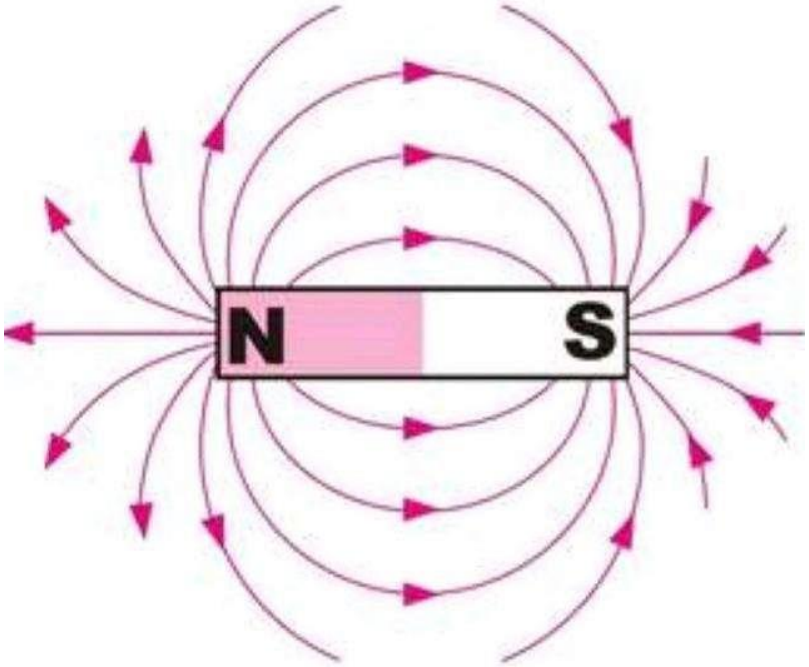
Solution: The right hand rule shows that the direction of the force is along the negative y-axis.

The magnitude of the force is:

$$F = qvB\sin\theta = (1.6 \times 10^{-19}\text{C})\cdot(3 \times 10^{5}\text{m}\cdot\text{s}^{-1})\cdot(2.0\text{T})\cdot(\sin 30^{\circ}) = 4.8 \times 10^{-14}\text{N}$$

7.1.3 Magnetic field lines and magnetic flux

We can represent magnetic field by field lines, just as the electric field.

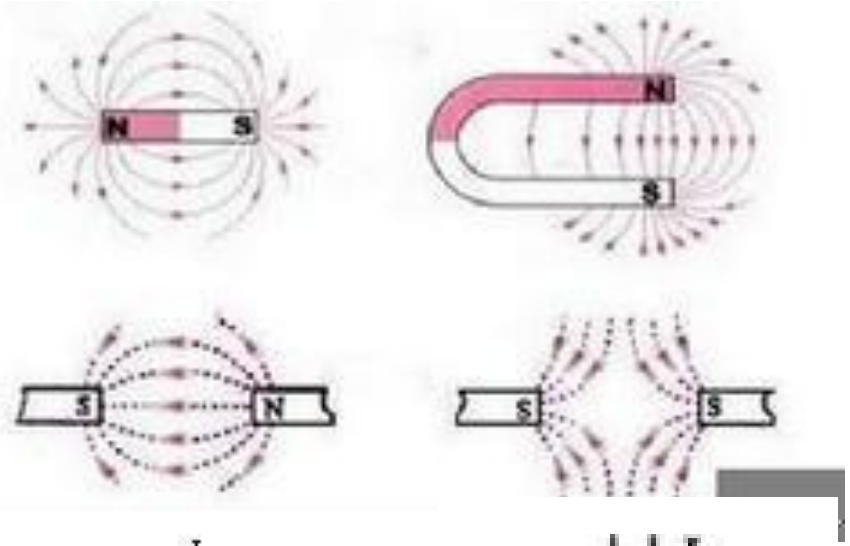


The same rules apply. That is, $\vec{B}$

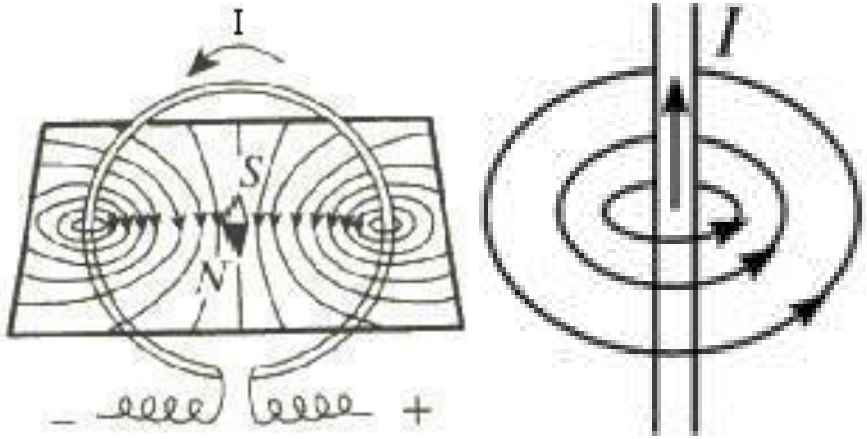
(1) Direction of the tangent: direction of

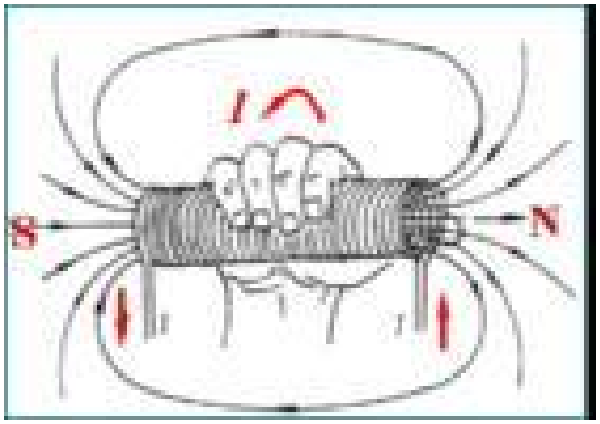
(2) Density of the magnetic lines: the magnitude of $\vec{B}$

7.1.3 Magnetic field lines and magnetic flux





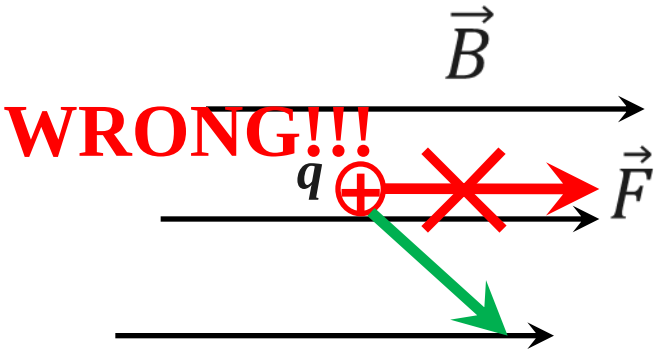


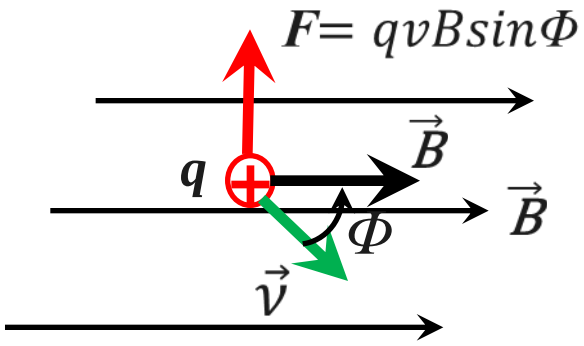


The magnetic field lines is closed or point to infinitude

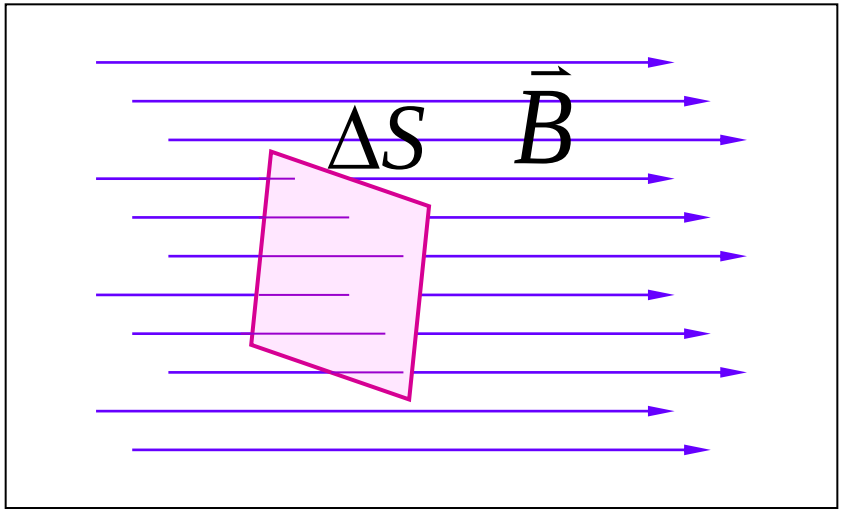
7.1.3 Magnetic field lines and related force

Note: Magnetic field lines are NOT “lines of force”. The force on a charged particle is **NOT** along the direction of a field line.





7.1.3 Magnetic field lines



Magnitude of magnetic field

$$B = \frac{\Delta N}{\Delta S}$$

ΔN: Magnetic field flux

ΔS: Area

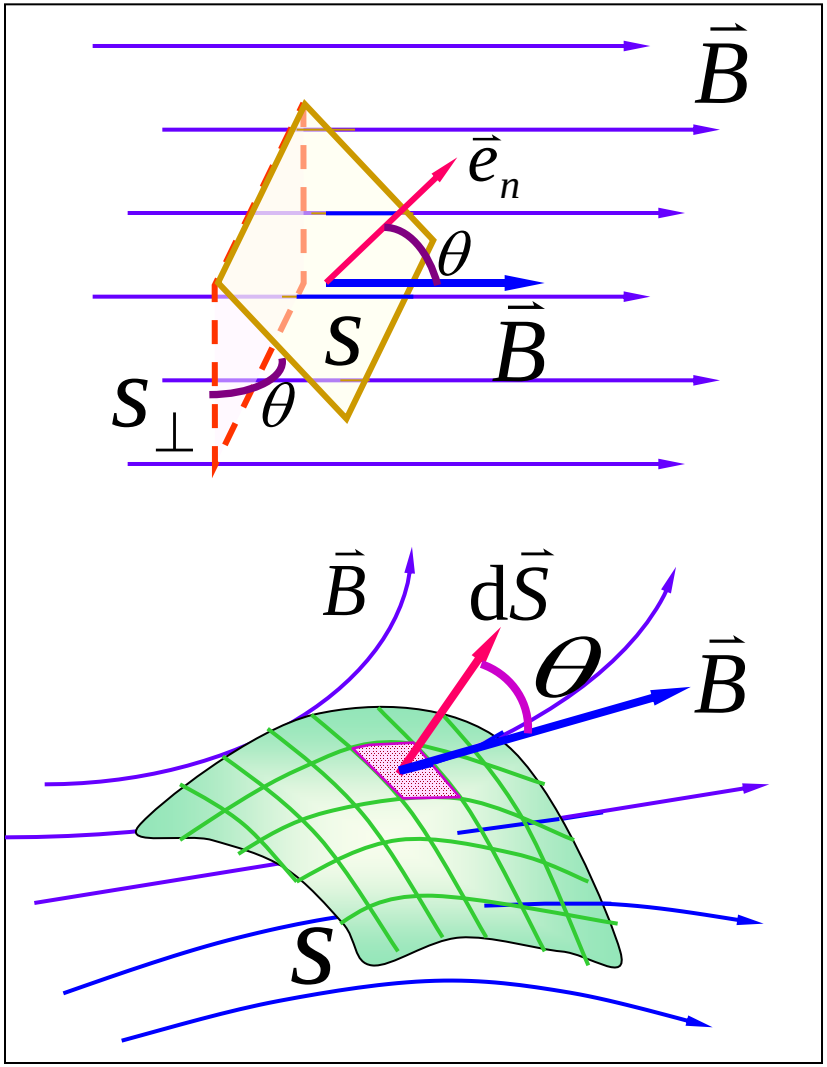
7.1.3 Magnetic flux

For any unit area,
the magnetic field is

$$d\Phi = B\cos\theta dS = B_n dS$$

General case

$$\Phi = \int_S \vec{B}\cdot d\vec{S}$$

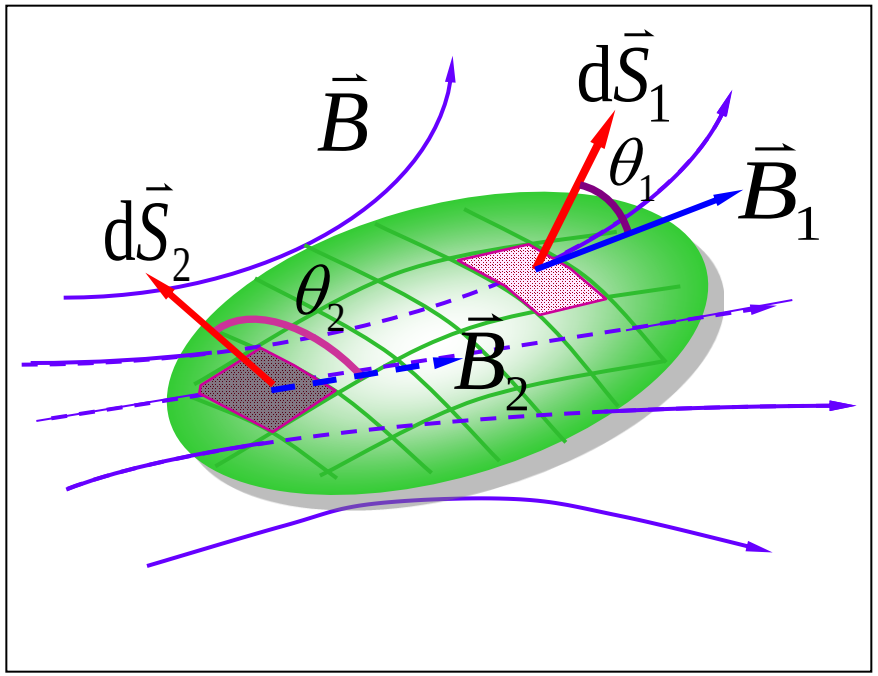


•Gauss law of magnetic field

The total magnetic field flux through a closed surface is always zero.

$$\oint_S B\cos\theta \mathrm{d}S = 0$$

$$\oint_S \vec{B}\cdot \mathrm{d}\vec{S} = 0$$



Magnetic field is vortex field

7.1.3 The Definition of Magnetic Field Magnitude

When the direction of moving charge is perpendicular to the magnetic field, the force exerted on the charge is the largest. And we define the magnetic field intensity as

$$B = \frac{F_{\perp}}{q_0 v}$$

Its unit is Tesla (T): $1T = 1N \cdot C^{-1} \cdot S \cdot m^{-1} = 1N \cdot A^{-1} \cdot m^{-1}$

Another unit is Gauss (G): $1G = 10^{-4}T$

7.2 Magnetic Force

7.2.1 Motion of Charged Particles in a Magnetic Field

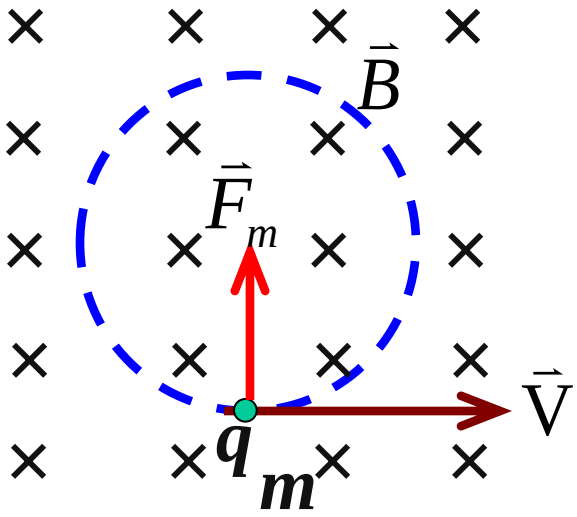
- **Uniform circular motion of charged particles in magnetic field**

$$F = qvB = ma = \frac{mv^2}{R}$$

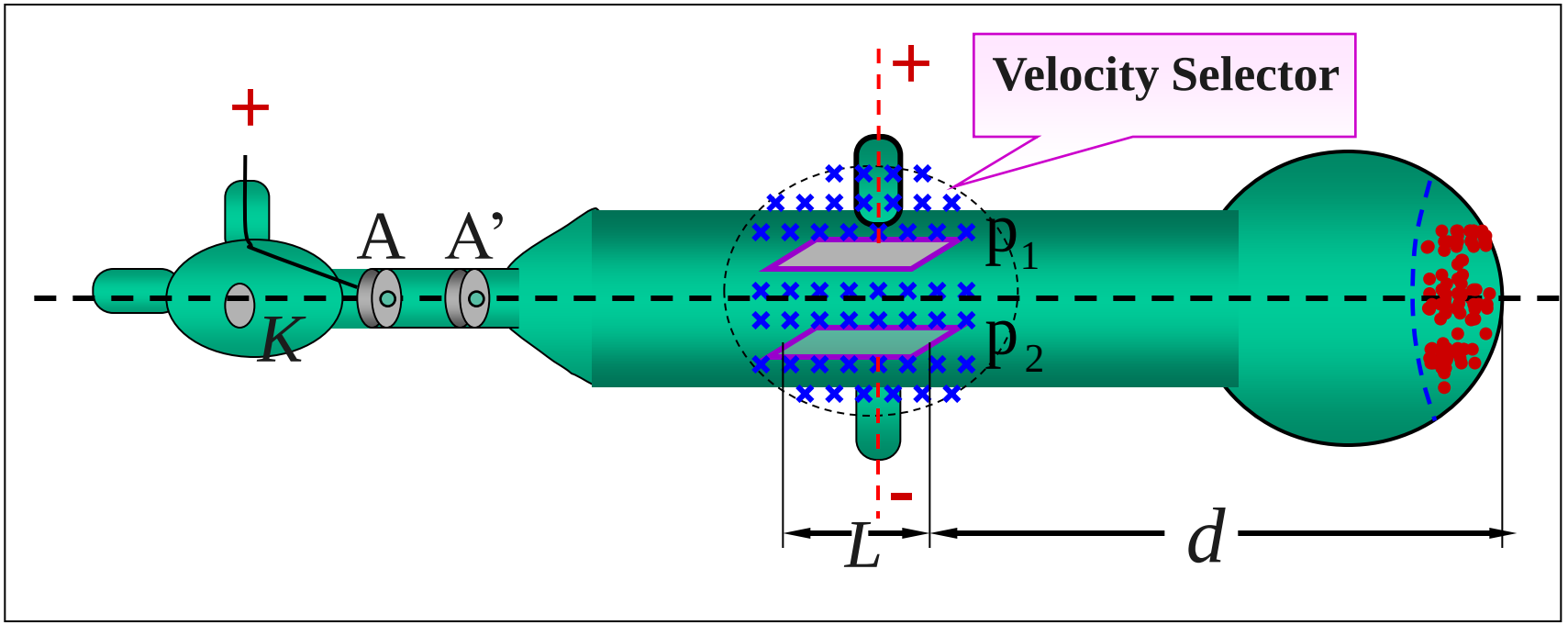
$$a = \frac{v^2}{R}$$ a: centripetal acceleration

Radius $$R = \frac{mv}{qB}$$

period $$T = \frac{2\pi R}{v} = \frac{2\pi m}{qB}$$

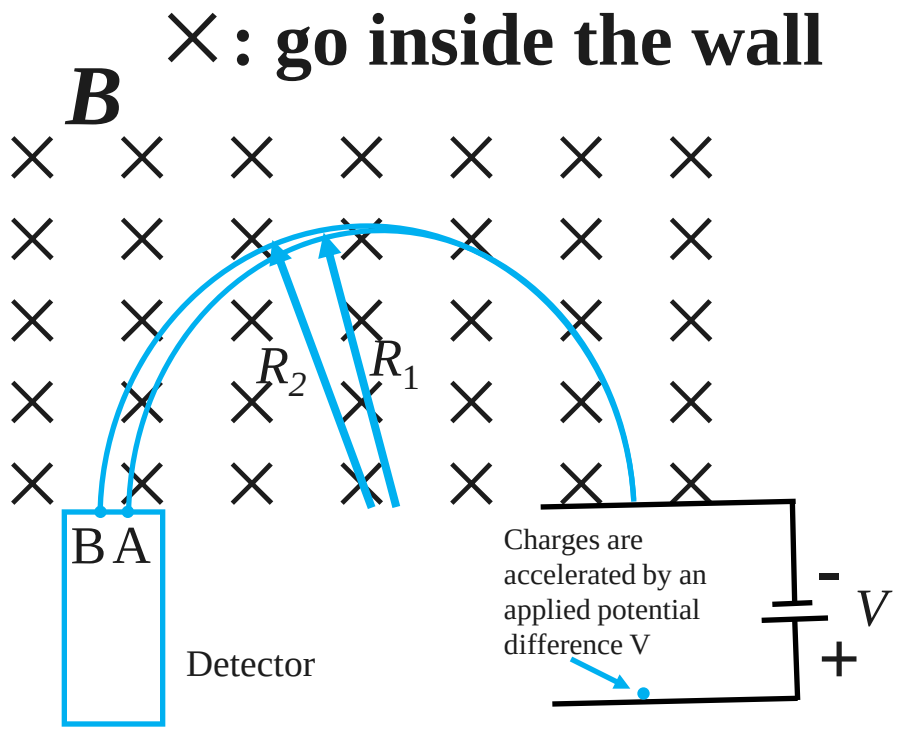


7.2.2 Velocity Selector and Mass Spectrometer



$$eE_0 = ev_0 \times B_0 \qquad v_0 = \frac{E_0}{B_0}$$

7.2.2 Mass Spectrometer



Mass spectrometer

Positively charged particles from the source are accelerated by an applied potential difference and enter a uniform magnetic field. The particles move in a semicircle of a radius r and then strike the detector.

In the electric field:

Work-energy theorem:

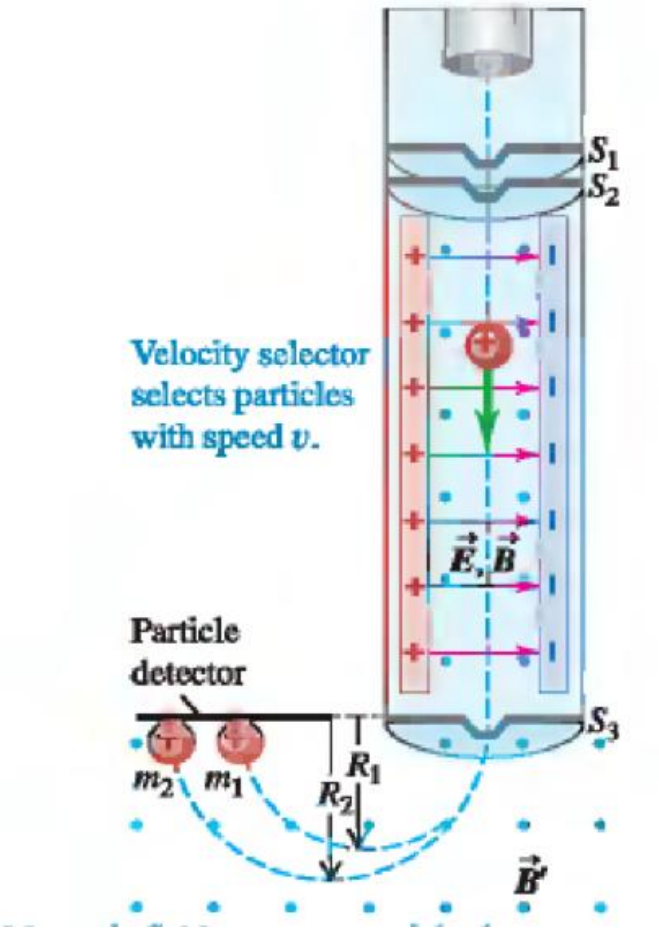
$$qV = \frac{1}{2}mv^2$$

$$R = \frac{mv}{qB}$$

$$\frac{m}{q} = \frac{B^2R^2}{2V}$$

$$\frac{m_1}{m_2} = \left(\frac{R_1}{R_2}\right)^2$$

7.2.2 Mass Spectrometer



Magnetic field separates particles by mass; the greater a particle's mass, the larger is the radius of its path.

Bainbridge's mass spectrometer

$$\frac{m_1}{m_2} = \left(\frac{R_1}{R_2}\right)^2$$

×: go inside the wall

•: go outside the wall

⊙: go outside the wall

Homework

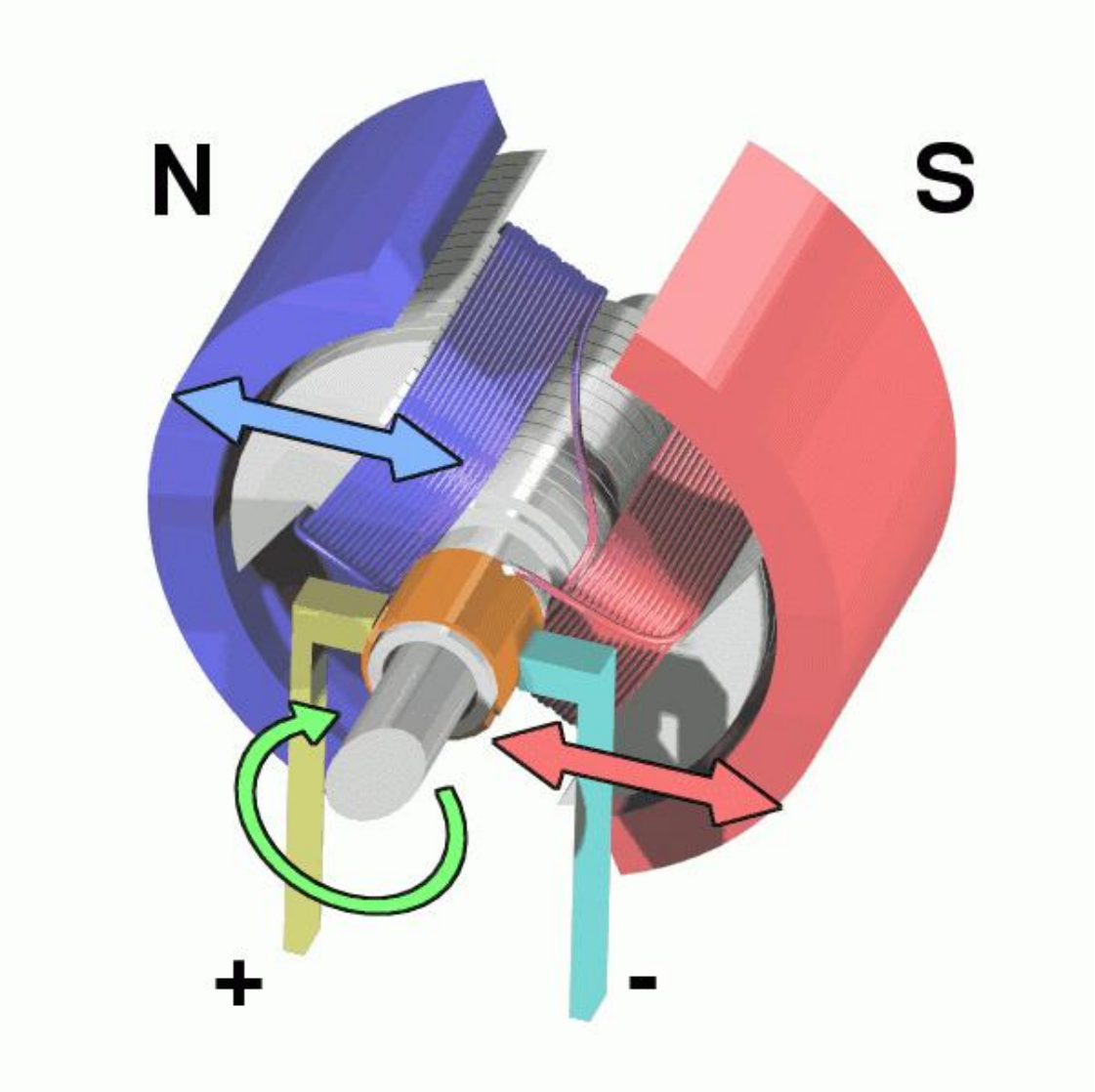
Page 161-162
Q1, 3, 4, 6 (week 1)

Homework file name:
Roll number+week X + name,
for example: 2023800901XX + week 1 + Alfa Robot

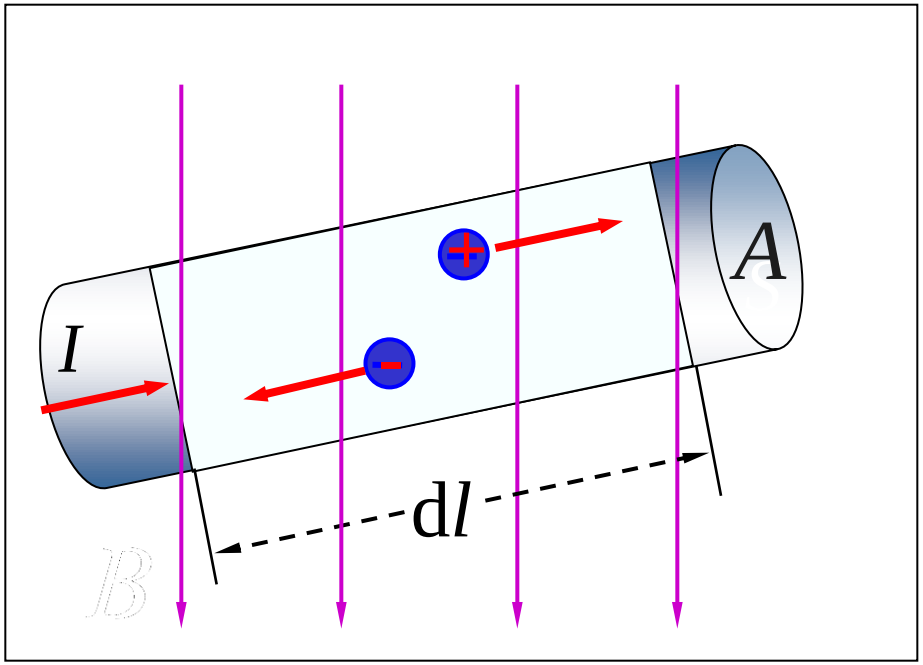
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202380090148 Golam Rabbi for Software Engineering),

7.2.3 The magnetic forces on current



7.2.3 The magnetic forces on current



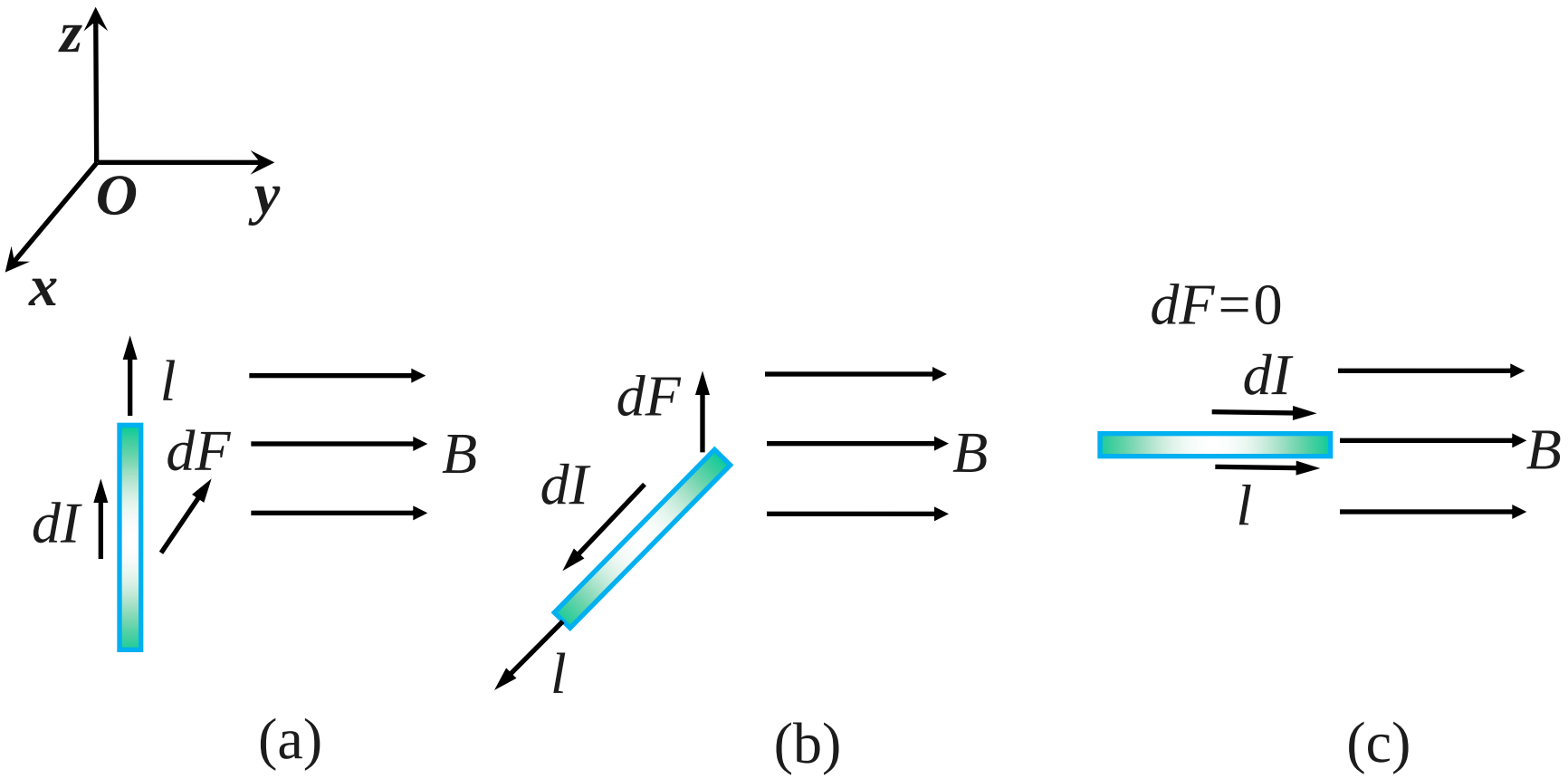
$$F = qv \times B = \mathrm{q}v\mathrm{B}\sin\theta$$

$$dF = (n_1 A dl)(q_1 v_1 B) + (n_2 A dl)(q_2 v_2 B)$$
$$= (n_1 q_1 v_1 + n_2 q_2 v_2) A dl B = J A dl B = I dl B$$

$$J = nqv$$

$$d\vec{F} = Id\vec{l} \times \vec{B}$$

7.2.3 The magnetic forces on current



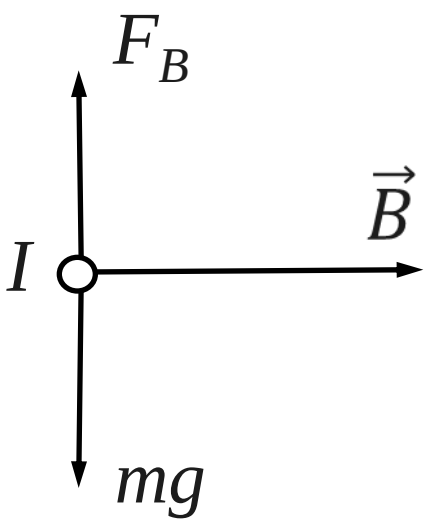
A wire segment, aligned with the (a) z-axis, (b) x-axis and (c) y-axis, in a magnetic field, with the force dF that act on the segment.

7.2.3 The magnetic forces on current

$$dF = Idl \times B$$

$$F = I \int \mathbf{d}l \times B$$

An example



A wire can be made to float in a magnetic field. The current in the wire from inner to outer

A straight, horizontal stretch of copper wire carries a current J = 28A. What are the magnitude and direction of the magnetic field needed to “float” the wire, that is, to balance its weight? Its linear density is 46.6g • m^{-1}.

Solution

Solution: For a length L of wire we have

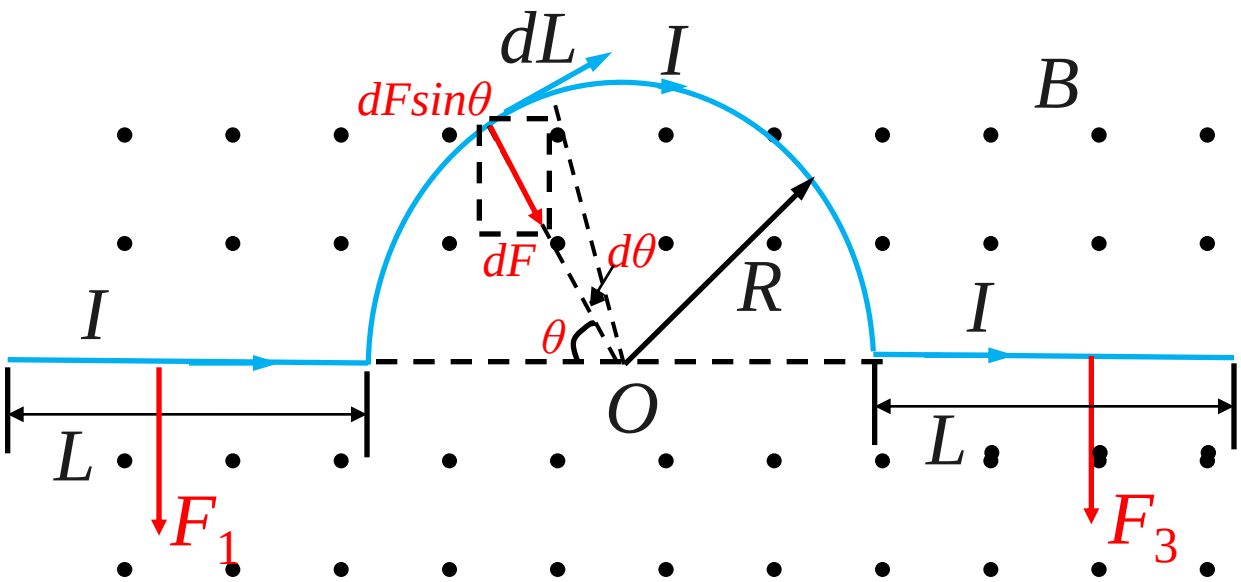
$$mg = LIB$$

$$B = \frac{(m/L)}{I} g$$

$$= \frac{(46.6 \times 10^{-3}\,\mathrm{kg \cdot m^{-1}})(9.8\,\mathrm{m \cdot s^{-2}})}{28\,\mathrm{A}}$$

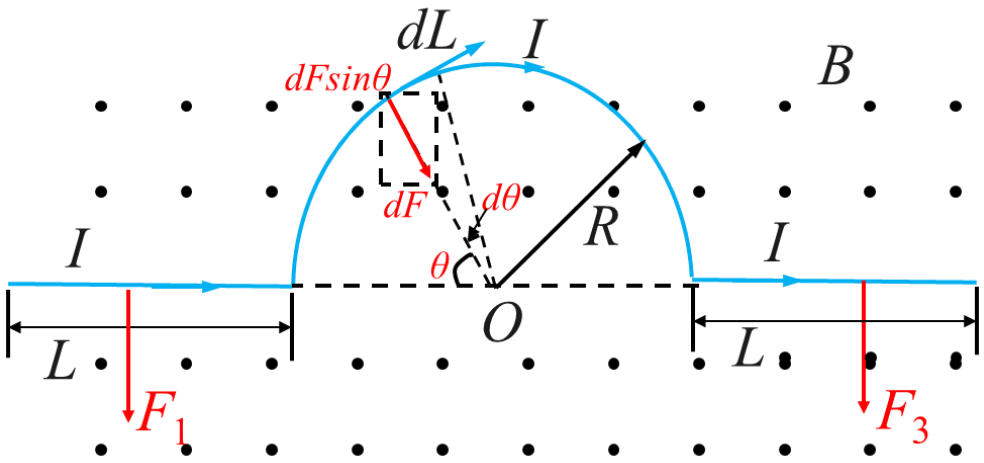
$$= 1.6 \times 10^{-2}\,\mathrm{T}$$

An Example



A wire segment, placed in a uniform magnetic field *B* that points out of the plane of the figure. If the segment carries a current *I*, what resultant magnetic force *F* acts on it?

Solution



The force that acts on each straight section

$F_1 = F_3 = ILBsin90° = ILB$

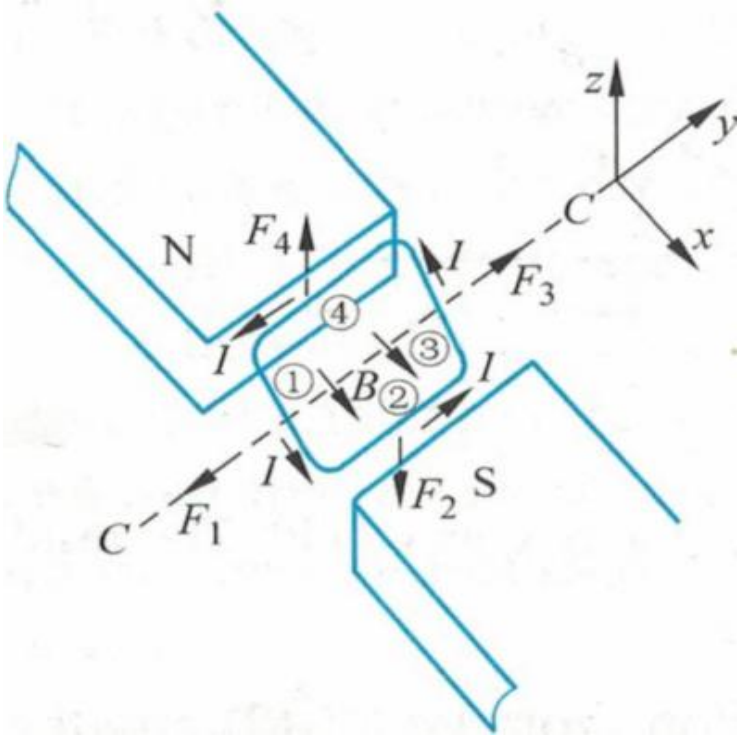
A segment of the arc of length dL has a force dF acting on it

$dF = IBdL = IB(Rd\theta)$

$$F_2 = \int_0^{\pi} dFsin\theta = \int_0^{\pi} IBRsin\theta d\theta = IBR\int_0^{\pi} sin\theta d\theta = 2IBR$$

$F=F_1+F_2+F_3$ = ILB + 2IBR + ILB = 2IB (L + R)

7.2.4 Torque on a current loop



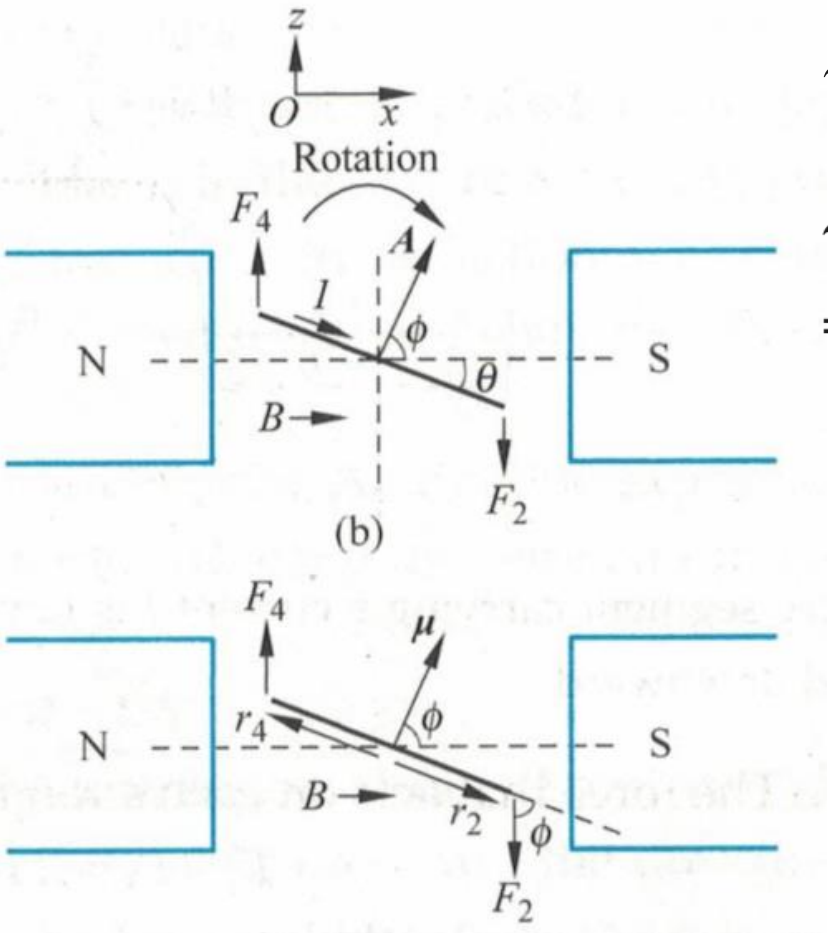
$F_1 = IaB\sin\theta$, in the $-y$ direction.

$F_2 = IbB$, in the $-z$ direction.

$F_3 = IaB\sin\theta$, in the $+y$ direction.

$F_4 = IbB$, in the $+z$ direction.

7.2.4 Torque on a current loop



The torque

$\tau = Fr$

$$\tau = \frac{a}{2}(IbB)sin\varphi + \frac{a}{2}(IbB)sin\varphi$$

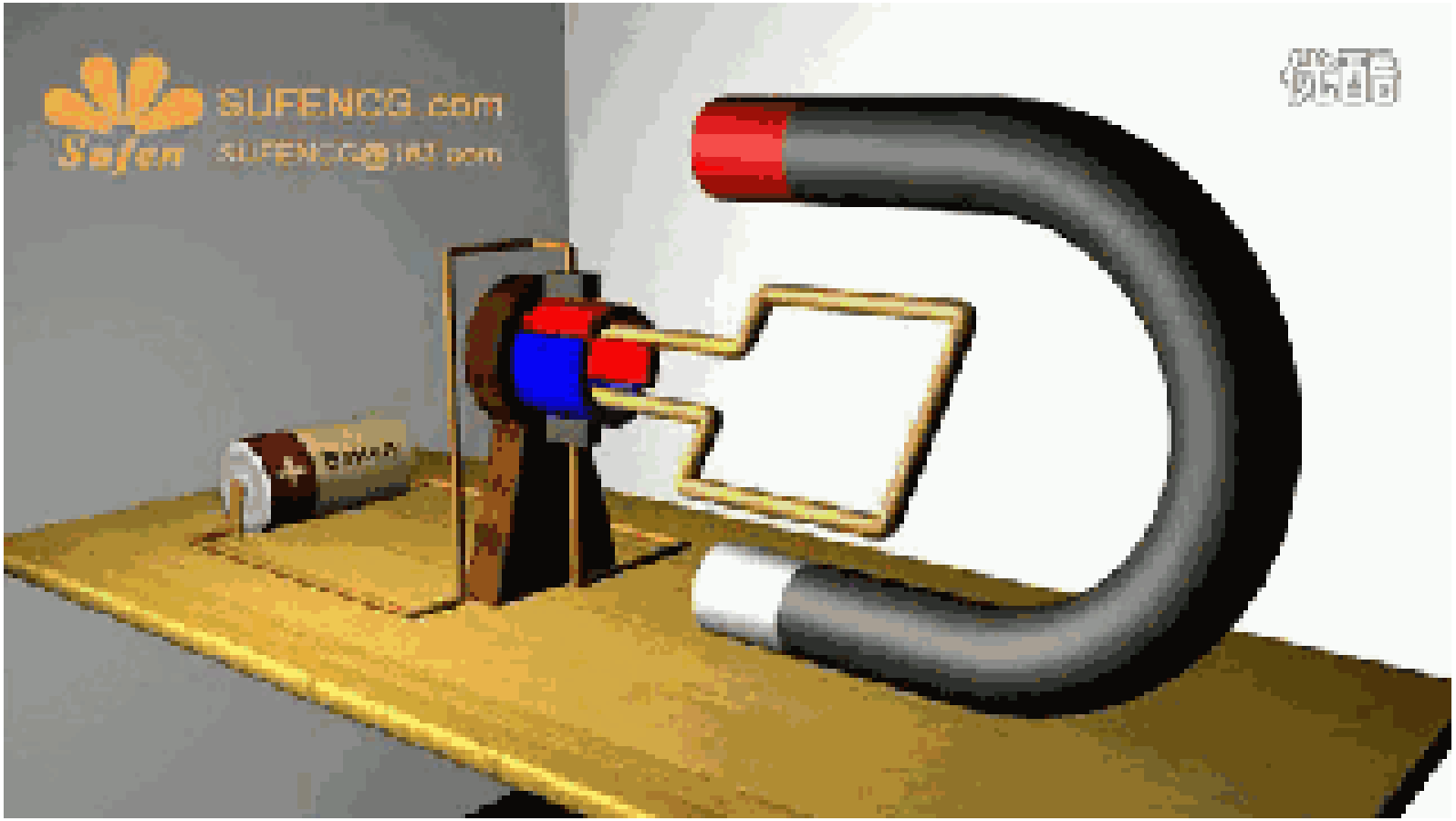
$= IabBsin\varphi$

$\tau = IABsin\varphi$ A: area of the loop, a*b

$\mu = IA$: magnetic dipole moment of the loop:

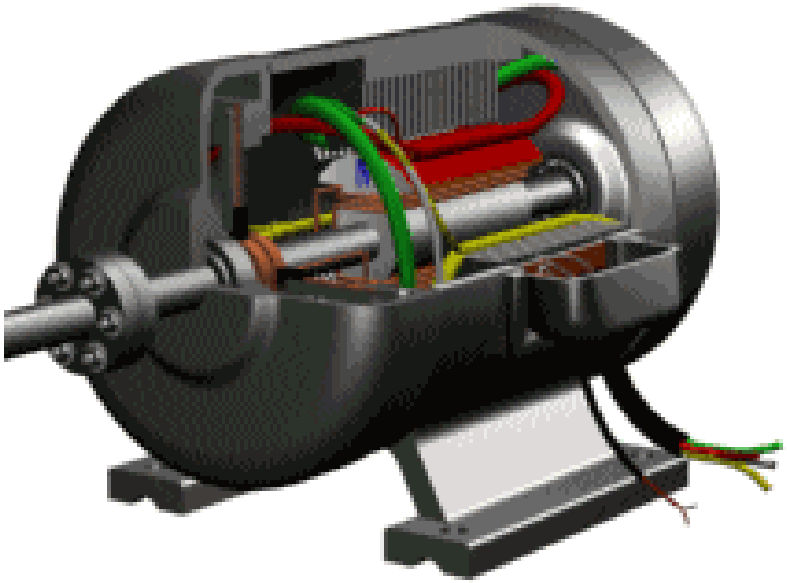
$\tau = \mu \times B = \mu Bsin\varphi$

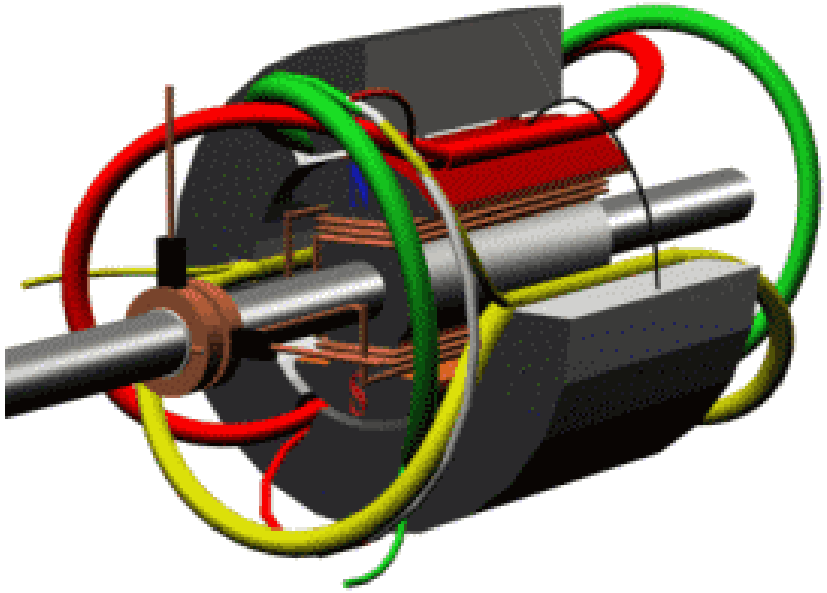
7.2.4 Electromotor



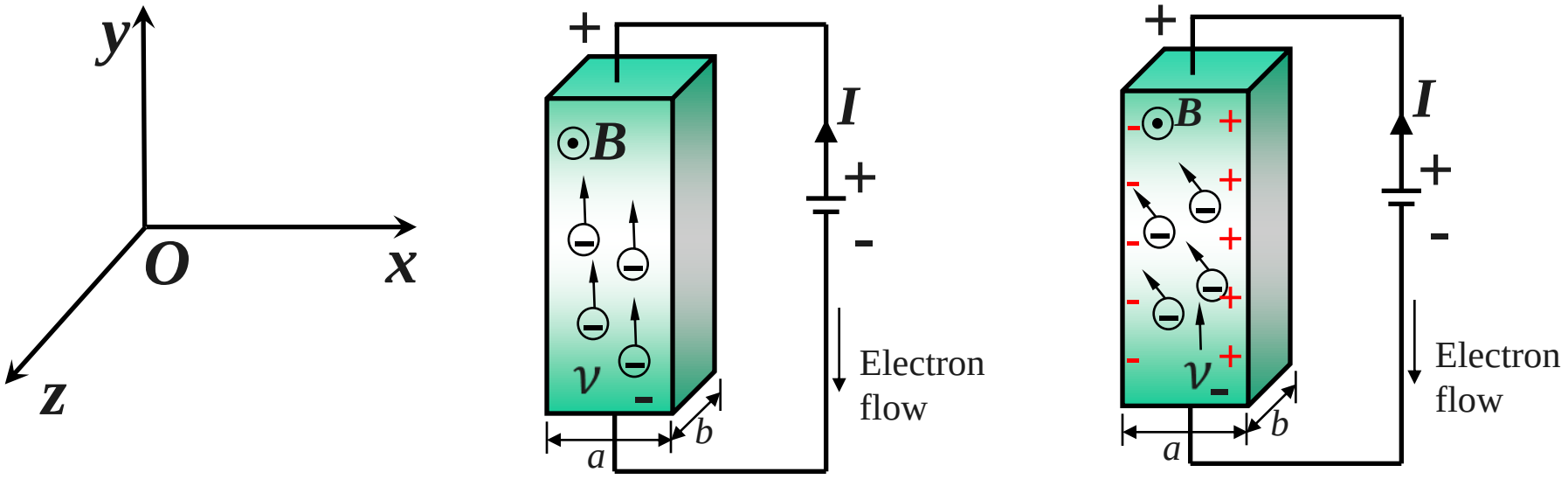
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7.2.4 Electromotor



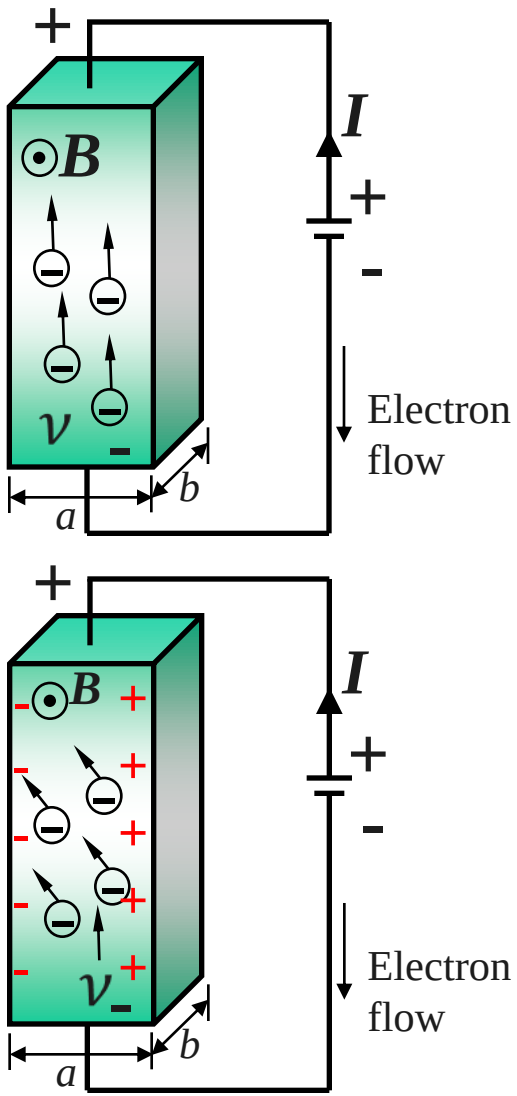


7.2.5 Hall Effect



The **Hall effect** is a potential difference perpendicular to the direction of current in a conductor, when the conductor is placed in a magnetic field.
The Hall potential is determined by the requirement that the associated electric field must just balance die magnetic force on a moving charge.
Hall-effect measurement can be used to determine the **sign of charge carriers** and **concentration**.

7.2.5 Hall Effect



When charge accumulate

$$eE_H = evB$$

$$E_H = \frac{V_H}{a}$$

V_H: Hall voltage, voltage formed by charge

a: width of strip

$$J = \frac{I}{ab}$$

$$v = \frac{J}{ne} = \frac{I}{abne}$$

$$V_H = \frac{I}{bne}B$$

Example

A strip of copper 150 μm thick is placed in magnetic field B = 0.65T and a current I = 23A is set up in the strip. What Hall voltage V_H will appear across the width of the strip? (The number of charge carriers per unit volume for copper is n = 8.47 × l0^{28} electrons • m^{-3})

Solution

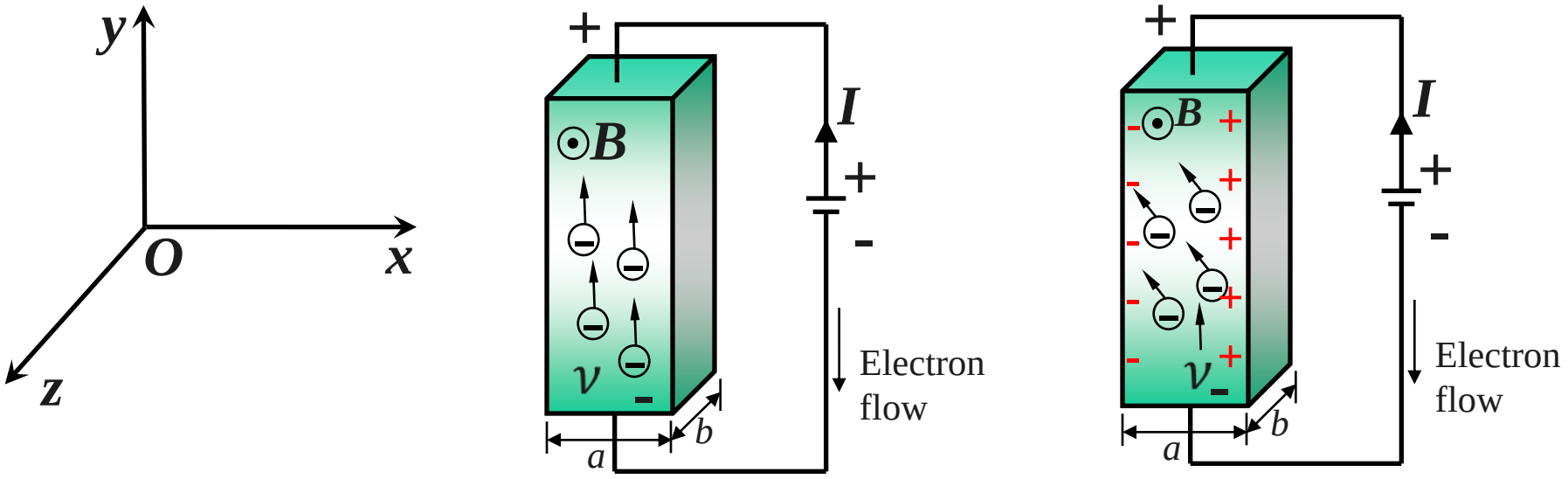
$$V_H = \frac{I}{bne} B$$

$$= \frac{(23\ \text{A})(0.65\text{T})}{(150\times10^{-6}\text{m})(8.47\times10^{28}\text{m}^{-3})(1.6\times10^{-19}\text{C})}$$

$$=7.4\times\ 10^{-6}\text{V}$$

$$=7.4\mu\text{V}$$

Application of Hall effect



Determine the sign of charge carriers and their concentration.

For example, determine p or n type of a semiconductor

Homework

Page 161-162

Q7, 9, 13 (week 3)

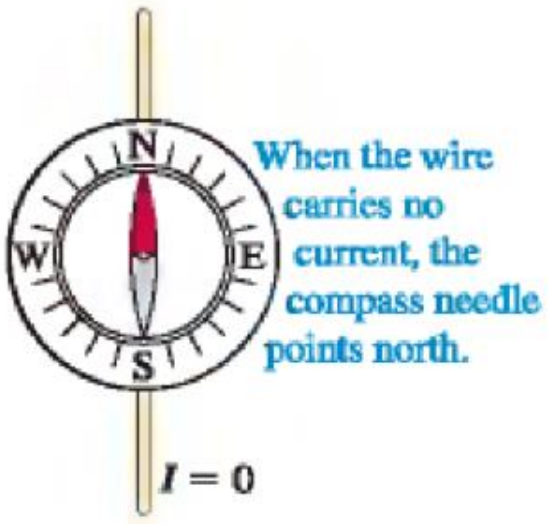
Homework file name:
Roll number+week X + name,
for example: 202380090001 + week 3 + Alfa

pdf file, send to monitors (Iben Walid Lilid – 202380040119 for Civil Engineering

202380090148 Golam Rabbi for Software Engineering)

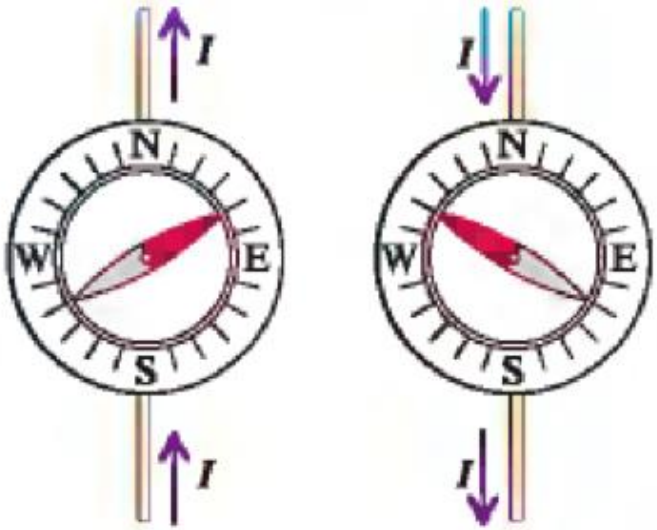
7.3 The property of Magnetic Fields

No current



A compass is placed over a horizontal wire without current

With current



The compass is deflected when there is a current. When the current reversed, the compass deflection is also reversed.

7.3 The property of Magnetic Fields

Ampere's law

$$\int B \cdot dl = \mu_0 I$$

$\mu_0 = 4\pi \times 10^{-7}$ Wb A^{-1} m^{-1} = $4\pi \times 10^{-7}$ T A^{-1} m

μ_0 is a constant, permeability constant of vacuum

The relationship between the magnetic field and the net current.

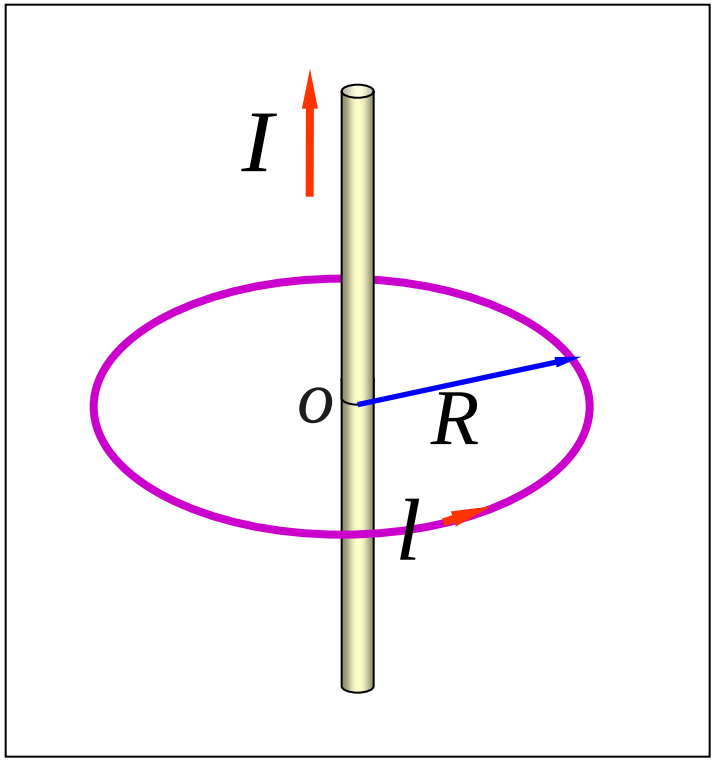
Ampere found that the lien integral of the magnetic field B around any closed path is equal to μ_0 times the net current across the area bounded by the path (or enclosed by the path)

7.3 Ampere's Law

$$\int B \cdot dl = \mu_0 I$$

$$B\, l \cos\theta = \mu_0 I$$

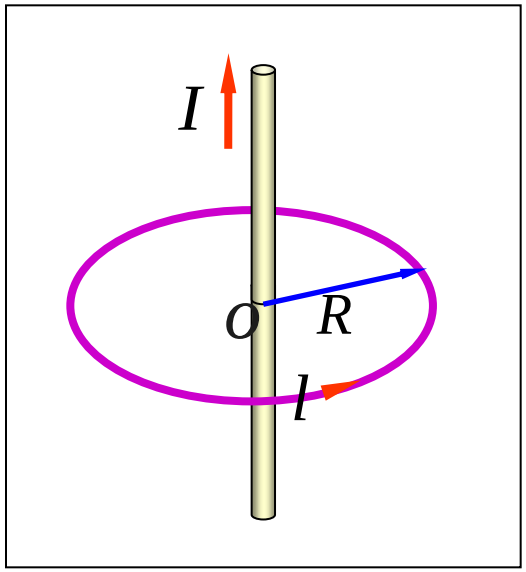
$$\oint_l \vec{B} \cdot d\vec{l} = \mu_0 I$$



l is a closed loop

Suggest l with I is right hand rule

7.3.1 example 1



Select a integration path:
A circle with the radius of R
Center is the current

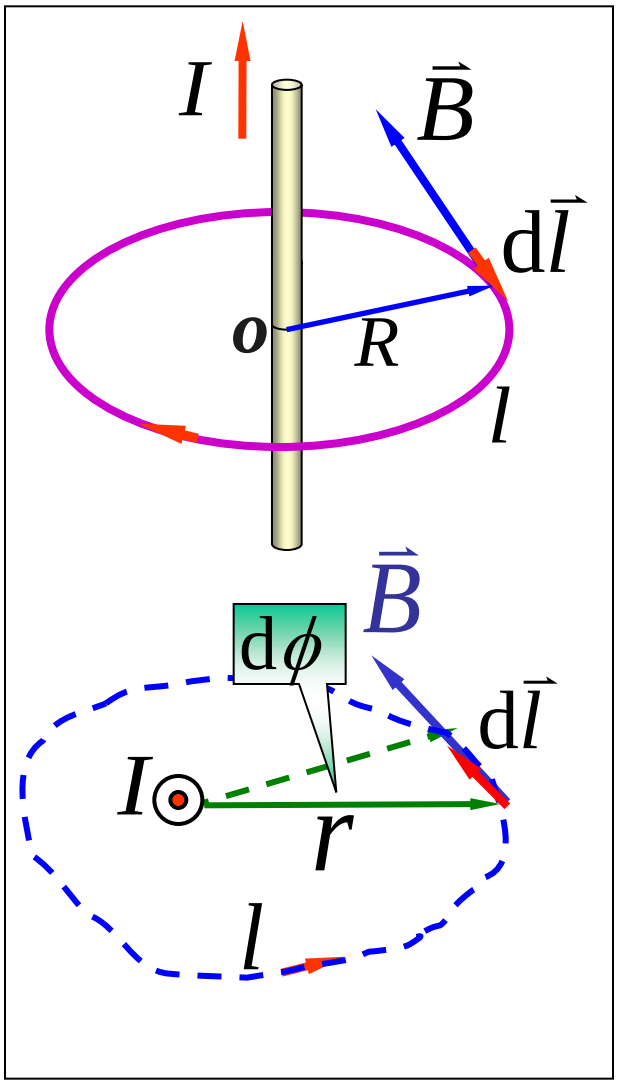
$$\int B \cdot dl = \mu_0 I$$

$$B \oint dl = \mu_0 I$$

$$B\ 2\pi R = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi R}$$

7.3 Ampere's Law



$$\oint_l \vec{B}\cdot \mathrm{d}\vec{l} = -\frac{\mu_0 I}{2\pi}\int_0^{2\pi} \mathrm{d}\phi = -\mu_0 I$$

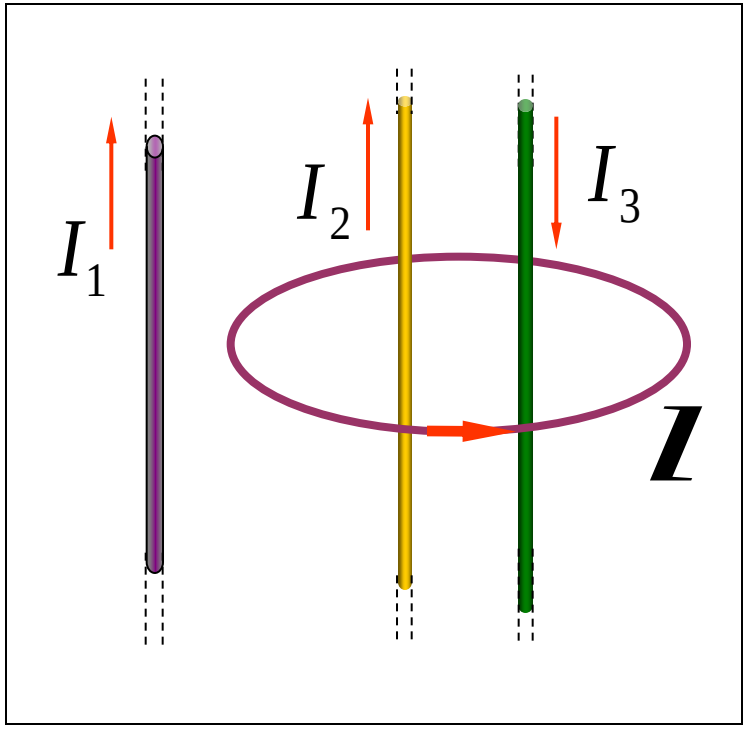
For arbitrary closed loop

$$\vec{B}\cdot \mathrm{d}\vec{l} = \frac{\mu_0 I}{2\pi r} r\mathrm{d}\phi = \frac{\mu_0 I}{2\pi}\mathrm{d}\phi$$

$$\oint_l \vec{B}\cdot \mathrm{d}\vec{l} = \mu_0 I$$

7.3 Ampere's Law

Multiple currents



$$\vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3$$

$$\oint_l \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 (I_2 - I_3)$$

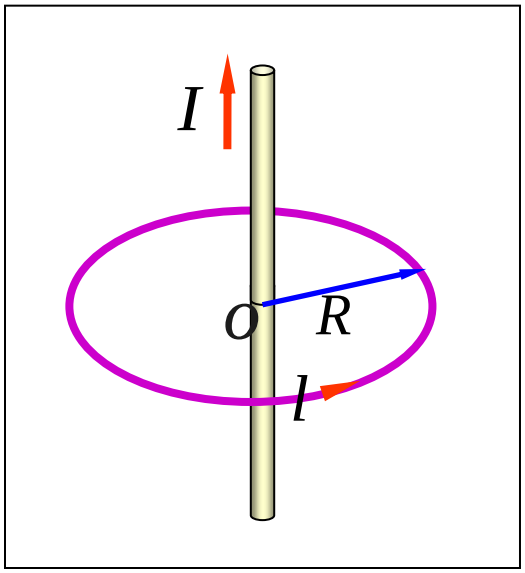
Ampere's Law for multiple currents

$$\oint \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 \sum_{i=1}^{n} I_i$$

7.3 Ampere's Law

$$\int B \cdot dl = \mu_0 I$$

$$B\, l \cos\theta = \mu_0 I$$



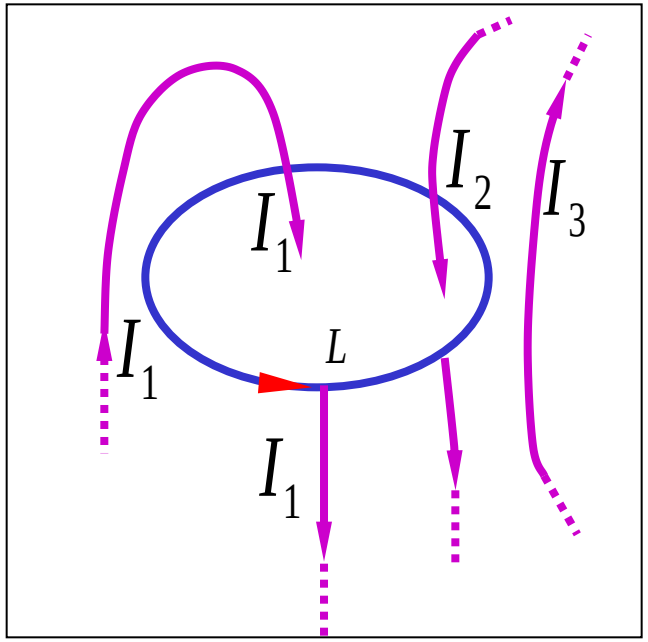
A few guiding principles:

1. If B is everywhere parallel to the path and has the same magnitude *B* at every point on the path, then its line integral is equal to *B* multiplied by the circumference of the path.

2. If *B* is everywhere perpendicular to the path, for all or some portion of the path, that portion of the path makes no contribution to the line integral.

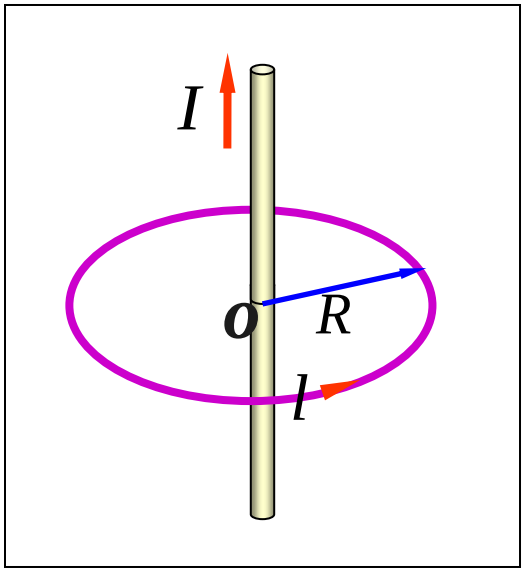
7.3 Ampere's Law

$$\int B \cdot dl = \mu_0 I$$



3. In the integral of magnetic field, B is always the total magnetic field at each point on the path. In general this field is caused partly by currents linked by the path and partly by currents outside. Even when no current is linked by the path, the field at points on the path need not be zero. In that case, however, the integration is always zero.

7.3: How to select integration path



4. Some judgments is required in choosing an integration path.

(a)point or points at which the field is to be determined must lie on the path,

(b)the path must have enough symmetry so that the integral can be evaluated.

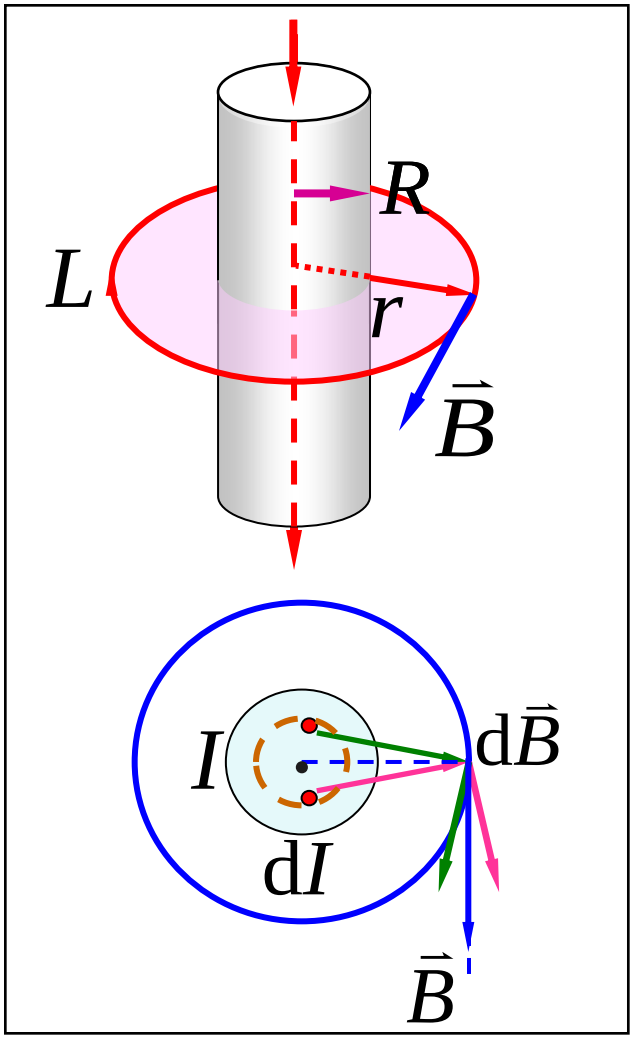
$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 \sum_i I_{in}$$

1）Constant current

2）Current is vector, means positive or negative. It is positive when I vs. l is with right hand rule

3）B is magnetic field contributed by all current. However, when do the calculation, only calculate the current inside the loop

7.3 example 1



The current, I within a wire whose cross section has radius R is known to be distributed uniformly over that cross section. What is the magnetic field as a function of the distance r from the wire's axis outside the wire and within the wire?

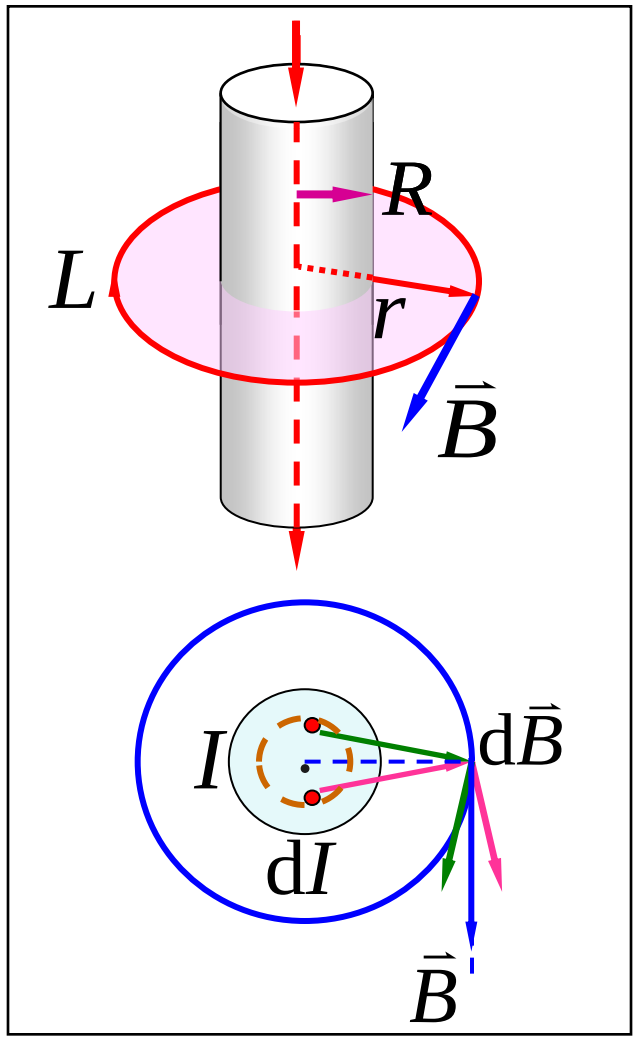
7.3 example 1: solution

Solution:

Step 1: select the loop

Step 2: $r > R$

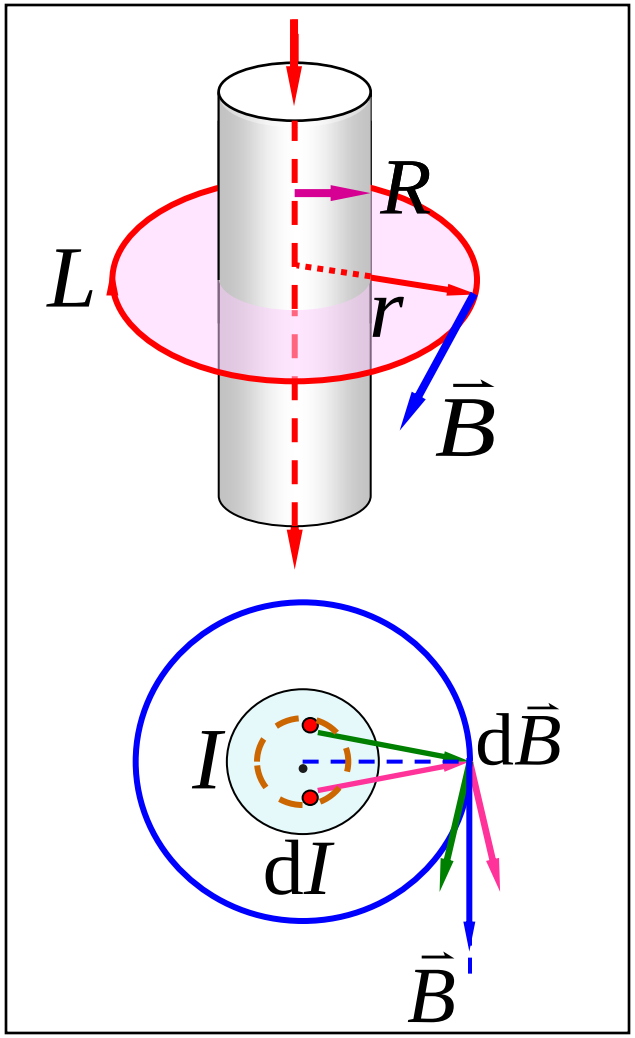
$$\oint_l \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 I \qquad B = \frac{\mu_0 I}{2\pi r}$$



7.3 example 1: solution

$$0 < r < R \qquad \oint_l \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 \frac{\pi r^2}{\pi R^2} I$$

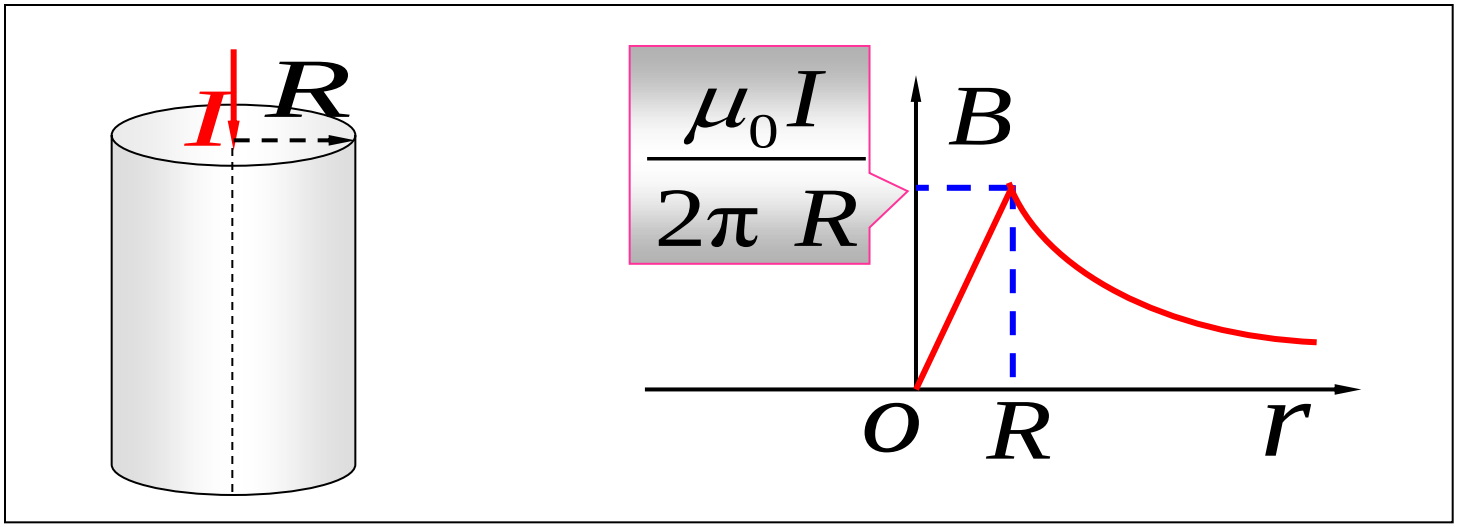
$$B = \frac{\mu_0 I r}{2\pi R^2}$$



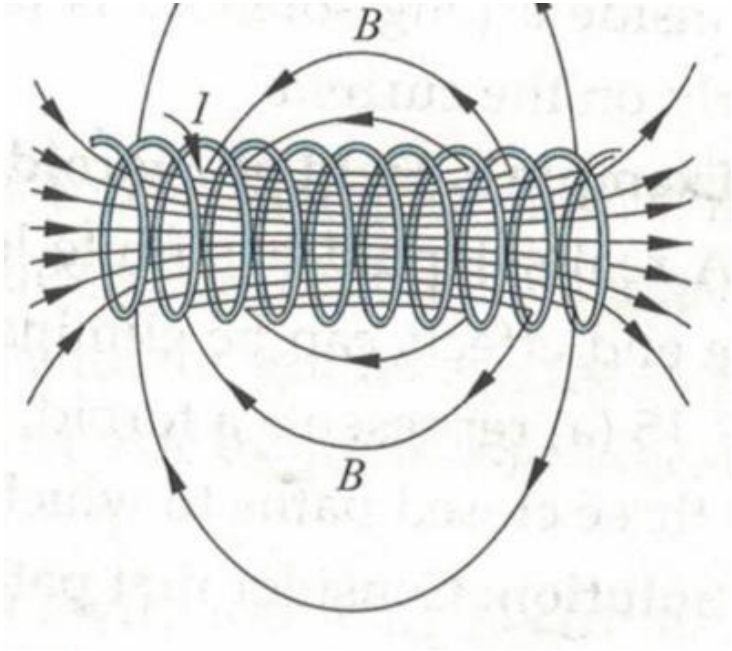
7.3 example 1: solution

$\vec{B}$ direction: right hand rule

$$\begin{cases} 0 < r < R, & B = \dfrac{\mu_0 I r}{2\pi R^2} \\ r > R, & B = \dfrac{\mu_0 I}{2\pi r} \end{cases}$$



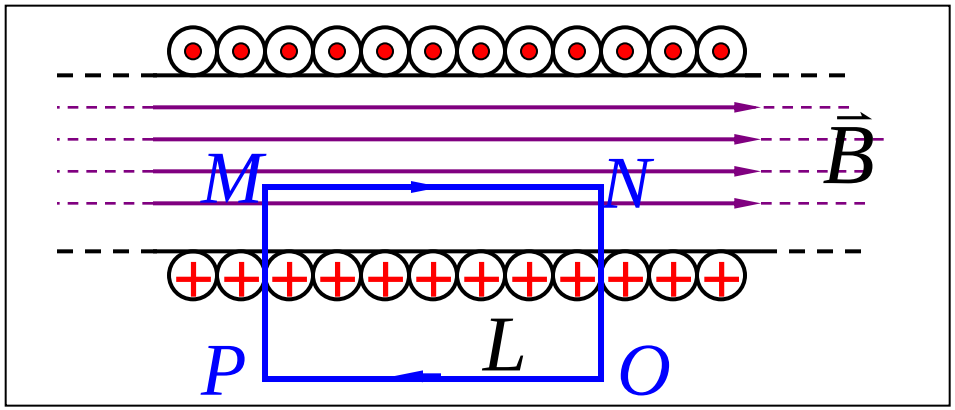
7.3 Example 3: field of a solenoid



A solenoid is an electrical element that generates magnetic fields similar to the electric fields generated by a parallel-plate capacitor. Just as a parallel plate capacitor sets up a constant electric field in space, so a solenoid generate a constant magnetic field in a region of space.

The magnetic field lines are densely spaced within a solenoid but widely spaced outside it. If the solenoid were infinitely long, the field outside would be zero.

7.3 example 3

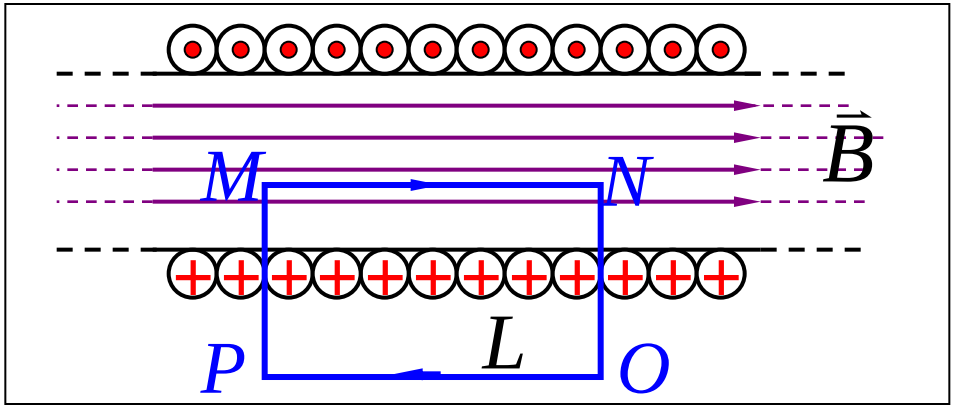


$$\oint_l \vec{B}\cdot \mathrm{d}\vec{l} = \int_{MN} \vec{B}\cdot \mathrm{d}\vec{l} + \int_{NO} \vec{B}\cdot \mathrm{d}\vec{l} + \int_{OP} \vec{B}\cdot \mathrm{d}\vec{l} + \int_{PM} \vec{B}\cdot \mathrm{d}\vec{l}$$

$$\int_{NO} \vec{B}\cdot \boldsymbol{d}\vec{l} = \mathbf{0} \qquad \int_{PO} \vec{B}\cdot \boldsymbol{d}\vec{l} = \mathbf{0} \qquad \int_{PM} \vec{B}\cdot \boldsymbol{d}\vec{l} = \mathbf{0}$$

$$\oint_{\boldsymbol{l}} \vec{B}\cdot \boldsymbol{d}\vec{l} = \int_{\boldsymbol{MN}} \vec{B}\cdot \boldsymbol{d}\vec{l}$$

7.3



$$\int B \cdot dl = \mu_0 I$$

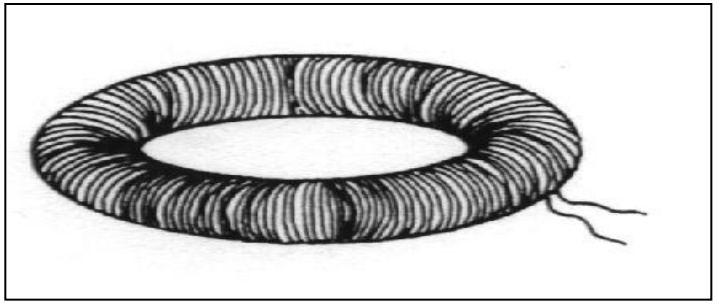
$$Bl = \mu_0 I_{enclosed} = \mu_0 \mathrm{N} \mathrm{I} = \mu_0 \mathrm{n} l \mathrm{I}$$

$$B = \mu_0 \mathrm{n} \mathrm{I}$$

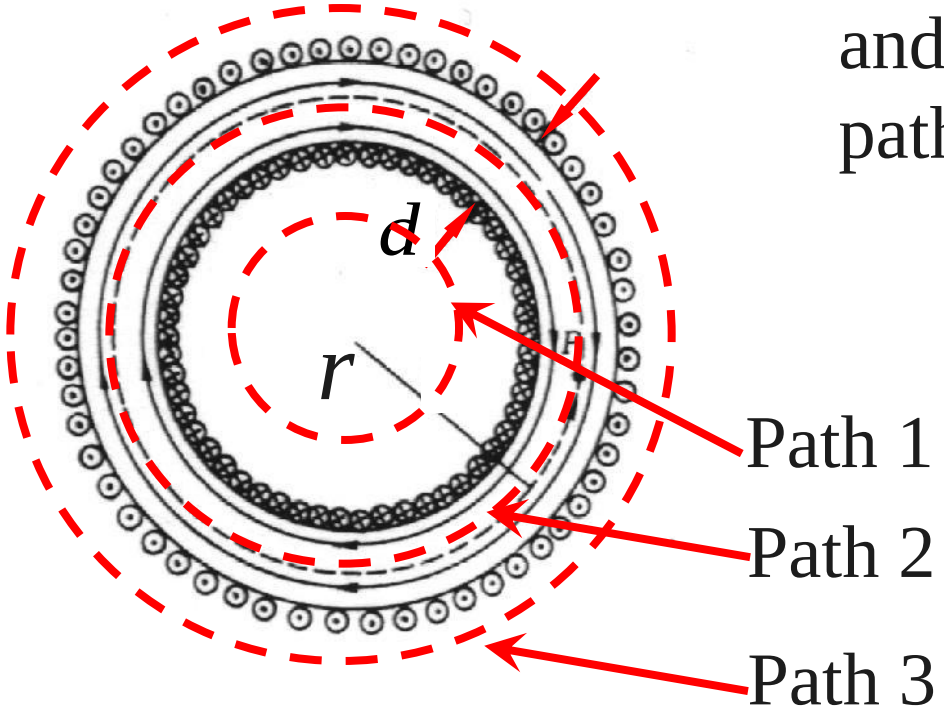
N: total number of wires passes the imaginary loop

n: wires density: wires unit length

7.3: Field of a toroidal solenoid (toroid)

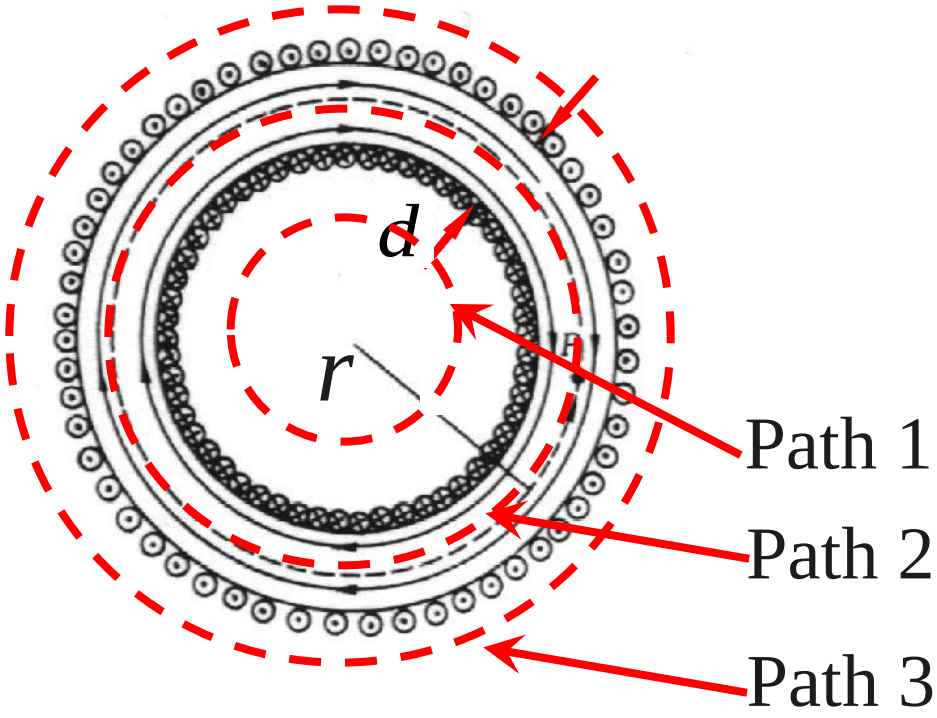


Doughnut shape, a toroid, wound with wire carrying a current I. Path 1, Path 2, and P 3 are three closed paths , calculate the B



7.3

Solution:
Path 1:



$$\int B \cdot dl = \mu_0 I$$

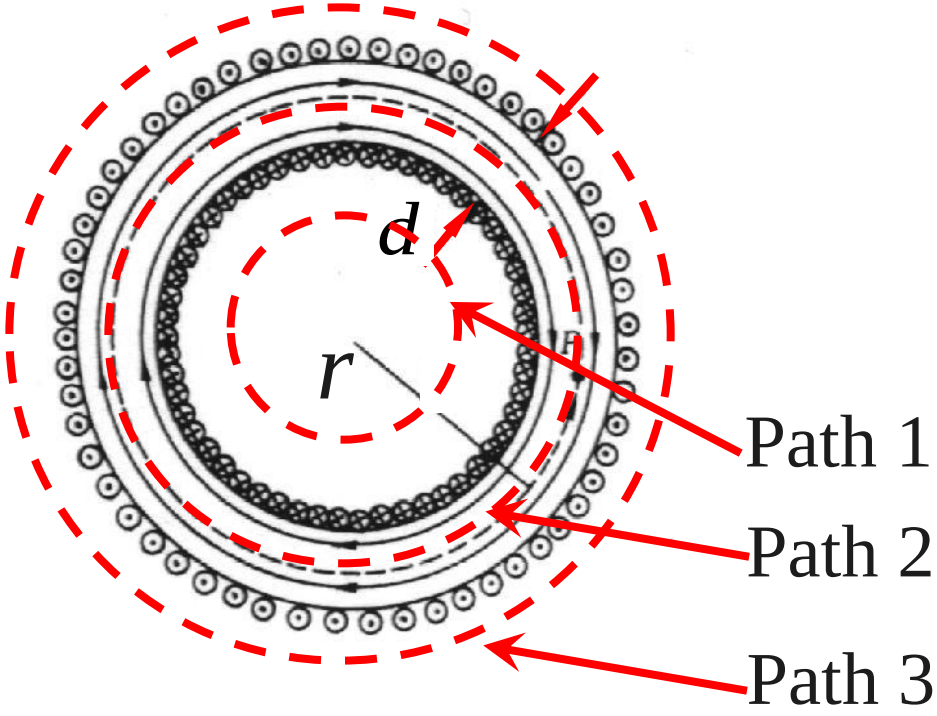
$$B\ 2\pi r = \mu_0 I = 0$$

Path 2:

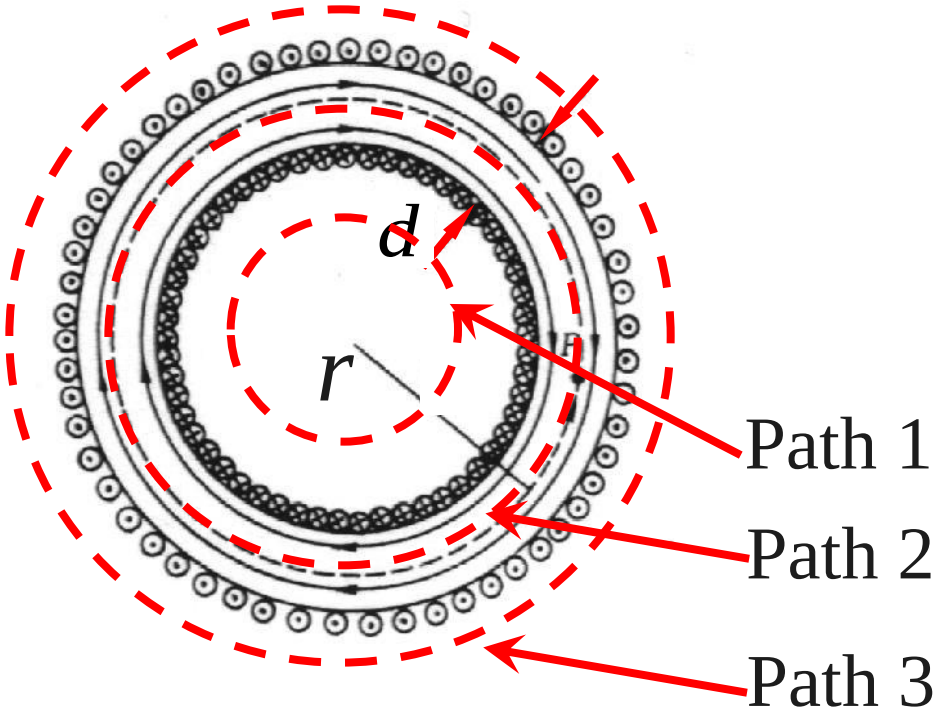
$$\int B \cdot dl = \mu_0 I$$

$$B\,2\pi r = \mu_0 N I$$

$$B = \frac{\mu_0 N I}{2\pi r}$$



Path 3:



$$\int B \cdot dl = \mu_0 I$$

$$B\ 2\pi r = \mu_0 (Iin + Iout) =$$

Questions

1. Of the three vectors in the equation $F = qv \times B$ which pairs are always at right angles? Which may have any angle between them?

2. Imagine that you are sitting in a room with your back to one wall and that an electro beam, traveling horizontally from the back wall toward the front wall, is deflected to your right .What is the direction of the uniform magnetic field that exists in the room?

Questions

3. If an electron is not deflected in passing through a certain region of space, can we be sure that there is no magnetic field in that region?
4. If a moving electron is deflected sideways in passing through a certain region of space, can we be sure that a magnetic field exists in that region?

Questions

5. If the magnetic field force does no work on a charged particle, how can it have effect on the particle's motion? Are there other examples of forces that do no work but have significant effect on a particle's motion?

6. Can a charged particle move through a magnetic field without experiencing any force? How?

Questions

8. A circular loop of wire lies on the floor of the room in which you are sitting. It carries constant current J in a clockwise direction, as viewed from above. What is the direction of magnetic dipole moment of this current loop?

9. Is it possible that, in a Hall effect experiment, no transverse potential difference will observed? Under what circumstances might this happen?

Questions

10. How could you use a magnetic field to distinguish between positive and negative particles ejected by radioactive nuclei?

11. A current was sent through a helical coil spring. The spring appeared to contract though it had been compressed. Why?

12. A proton moves parallel to a long straight wire, in the direction of the current carrier the wire. What is the direction of the force on the proton?



Homework

Page 161-162

Q14, 15 (week 4)

Homework file name:
Roll number+week 4 + name,
for example: 202380090001 + week 4 + Alfa

In pdf file, send to monitors (Iben Walid Lilid – 202380040119 for Civil Engineering

202380090148 Golam Rabbi for Software Engineering)

8V

7. A thin, straight wire carries a current of 10mA and makes an angle of 60° with a constant magnetic field of magnitude 10^{-4}T. The portion of the wire in this field has a length of 10cm. Calculate the force, both direction and magnitude, on this segment of the wire.

9. Fig. 7-19 shows a long wire carrying a current of 30A. The rectangular loop carries a current of 20 A. Calculate the resultant force acting on the loop. Assume that $a = 1.0\text{cm}$, $b = 8.0\text{cm}$, and $L = 30\text{cm}$.

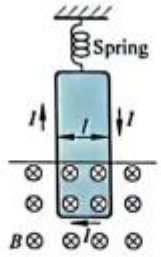


Fig. 7-18 Problem 8.

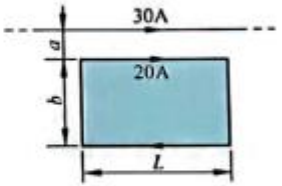


Fig. 7-19 Problem 9.

13. A Hall probe used to measure magnetic fields has a thickness of 1.5mm and is made from a material having a charge carrier density of $10^{23}m^{-3}$. For a current of 100mA, a Hall voltage of 0.42mV results when the probe is placed in a magnetic field of unknown strength. What is the value of the magnetic field? (The charge carrier is electron.)

14. The probe that demonstrates the Hall effect is used to measure the density of charge carriers in an unknown sample of metal. A sample of the material 0. 6mm thick is placed in a magnetic field of 0.50T. When a current of 100 mA passes through the material, a Hall voltage of 110μV is measured. What is the density of charge carriers? (The charge of the charge carriers is the same as electrons.)

15. A solenoid of diameter 3.0cm has a length of 60cm and 450 turns of wire. What is the magnetic field at the center of the solenoid when the current in the coil is 5A ? What is the magnetic flux through the cross-sectional area of the solenoid ?